TECHNICAL PROJECT: MULTI-VLAN ENTERPRISE NETWORK DEPLOYMENT

Environment: Cisco Packet Tracer

Network Engineer:
Pilusa Obed

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Quality of Service(QoS) Implementation

Implementing Quality of Service (QoS) is critical in this scenario especially considering the type of traffic involved and here's why:

- Types of traffic and their requirement: AMIGO LTD is handling various types of traffic which includes:
    - Voice over IP (VoIP) a real-time application that is sensitive to latency, jitter and packet loss.
    - Video Conferencing an application that requires consistent bandwidth, low latency and low jitter while being sensitive to buffering and frame drops.
    - Cloud-based CRM tools this are less sensitive to delay but require reliable throughput and minimal packet loss for productivity.
- Without QoS in place, these varying traffic types compete for bandwidth during peak time. This causes performance issues such as laggy calls, frozen video feeds, and slow-loading CRM pages. QoS helps by ensuring that delay-sensitive applications like VoIP and video get priority, improving user experience without needing to upgrade bandwidth.

This is how QoS mechanisms will improve performance for the affected applications.

- Traffic Classification with DSCP (Differentiated Services Code Point): It ensures that VoIP and video packets are marked and prioritized throughout the network path, reducing latency and jitter.
- Traffic Shaping: Smooths out traffic bursts by controlling flow, avoiding congestion, and giving real-time apps more stability.
- Bandwidth Allocating: Reserves minimum bandwidth for critical services like video conferencing, making sure they don't suffer during peak hours.
- By applying QoS rules at access switches or routers, delays can be reduced early on before congestion can affect performance.

Service Continuity & downtime Mitigation:

Here's a well-structured strategy that reduces downtime and avoids breaking services:

- Dual stack is a technique that deploys dual stack devices which are devices with network interfaces that can originate and understand both IPv4 and IPv6 packets as explained by (Juniper Networks , 2024). By applying this technique first, we ensure a gradual rollout and testing without interrupting IPv4 Traffic.
- Tunnelling: is a technique used to connect IPv6-enabled systems over an existing IPv4 infrastructure. This technique can be used in the middle of transitioning when the core network is IPv6-ready but there are still networks that rely on IPv4.
- Translation: is a technique used to enable communication between IPv4 and Ipv6 networks by translating one protocol to the other, as defined by (GeeksforGeeks, 2024). This can be used as the last technique to communicate with legacy systems critical for operation and may take them longer to transition.

Next-Generation IPv6 Routing:

My recommendations for suitable IPv6 routing protocols for the organization can be found below:

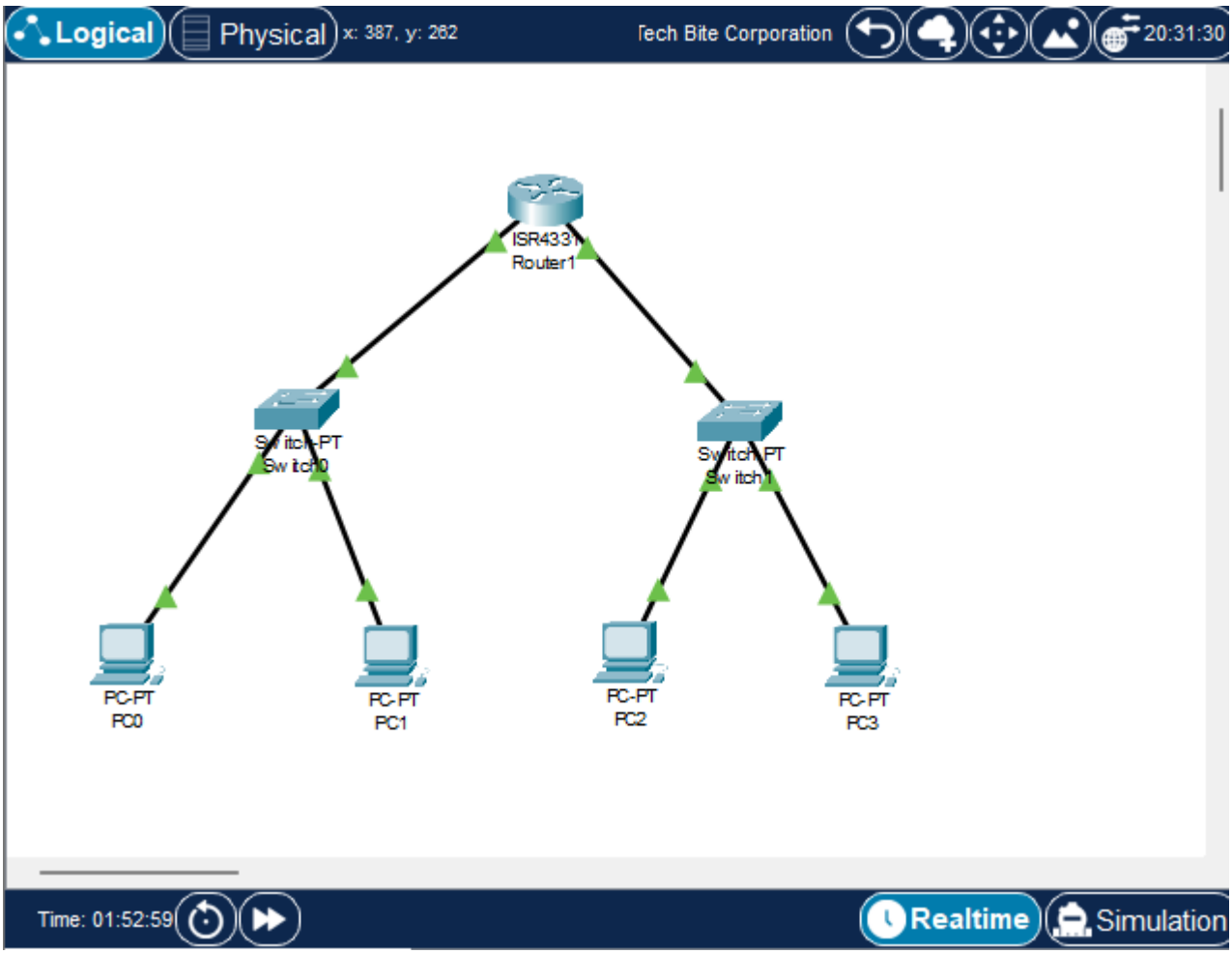
- OSPFv3(Open Shortest Path First for IPv6): This is an upgraded version of OSPF that fully supports IPv6 addressing. It's a link-state routing protocol, this means that it builds a map of the network topology and uses it's algorithm to determine the best route. OSPSv3 is a vendor-neutral protocol meaning it can be highly beneficial in a multi-vendor environment.
- EIGRP for IPv6(Enhanced Interior Gateway Routing Protocol): Is a protocol that combines the speed of distance vector protocols and the precision of link-state protocols making a hybrid routing protocol. It can be easily configured while providing redundancy by quickly recalculating routes if a link goas down.

Network Topology and Design(IPv6)

Using Packet Tracer to help Amigo Corporation with their network upgrade initiative by following the following steps:

a. Creating a topology using the following devices:
   i. 1 Router
   ii. 2 Switches
   iii. 4 PCs (2 connected to each swich)

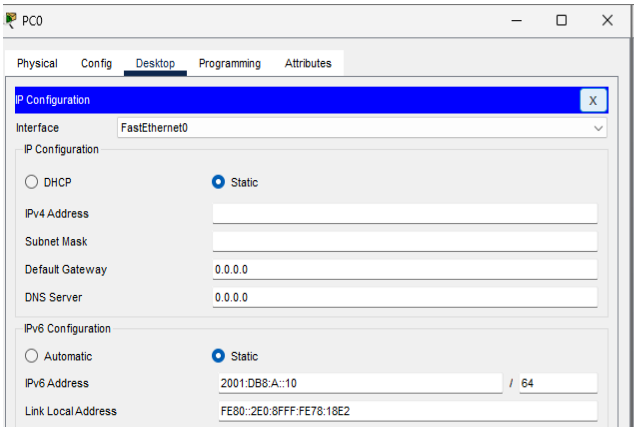
Find the Star topology:

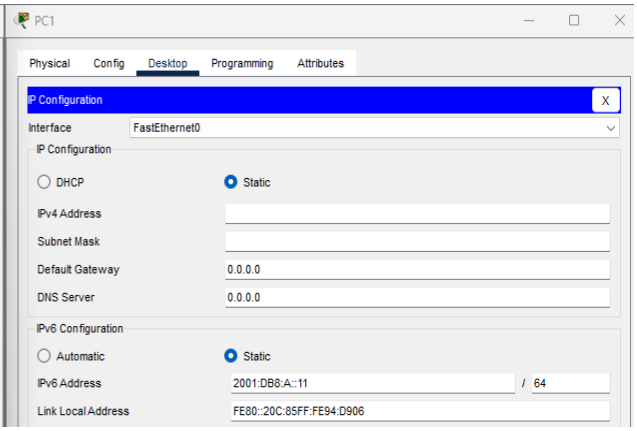


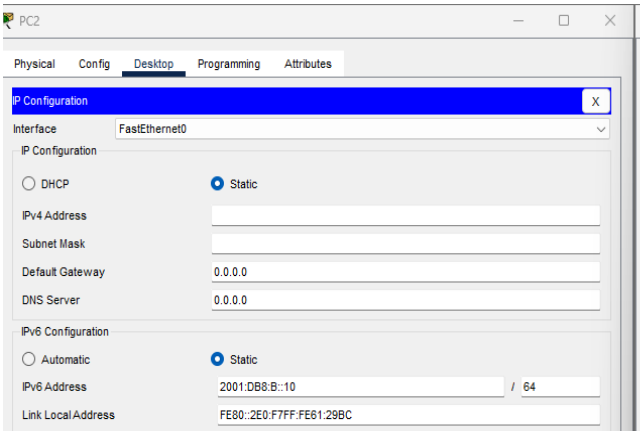
b. Assigning IPv6 Addresses.
   i. Network A (Left LAN): 2001:DB8:A::/64
   ii. Network B (Right LAN): 2001:DB8:B::/64
   iii. Router interfaces

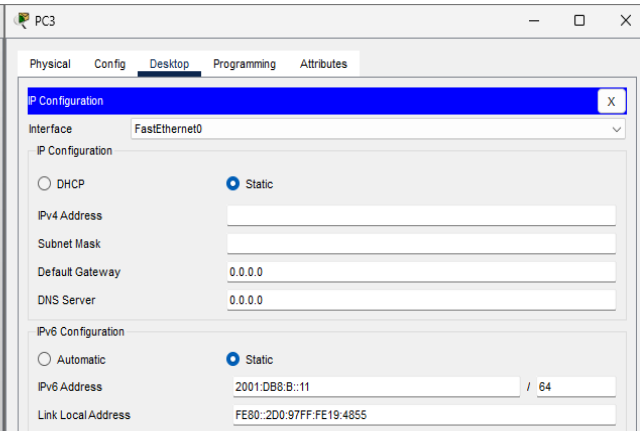
```
        --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: NO


Press RETURN to get started!



Router>EN
Router>ENable
Router#confi
Router#configure t
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ipv6 uni
Router(config)#ipv6 unicast-routing
Router(config)#interface
Router(config)#interface gi
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#ipv6 ad
Router(config-if)#ipv6 address
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ipv6 address
Router(config-if)#ipv6 address 2001:DB8:A::1/64
Router(config-if)#NO SH
Router(config-if)#NO SHutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed
state to up

Router(config-if)#
Router(config-if)#interface GigabitEthernet0/0/1
Router(config-if)#ipv6 address 2001:DB8:B::1/64
Router(config-if)#NO SHutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed
state to up
```

iv. PCs



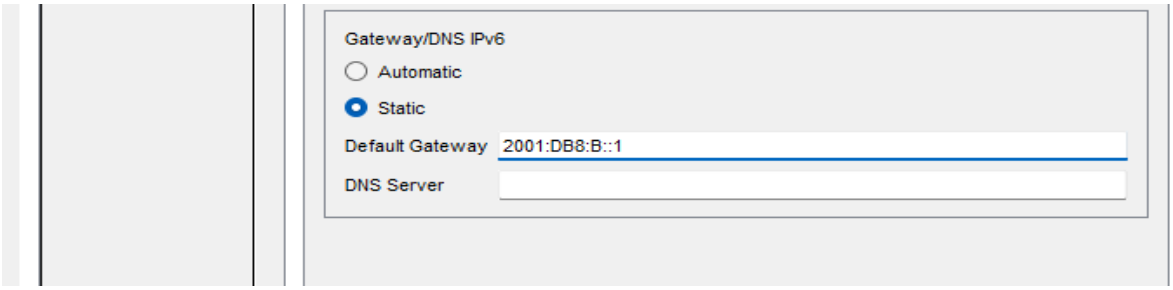




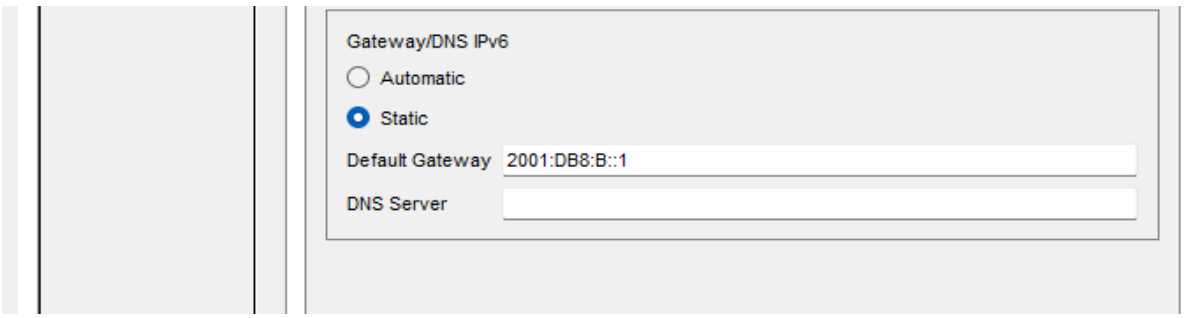


c. Setting the router's interface address as the default gateway on each PC

PC0 and PC1



PC3 and PC4



d. Configuring IPv6 addresses on Router Interfaces

```
        --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: NO


Press RETURN to get started!



Router>EN
Router>ENable
Router#confi
Router#configure t
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ipv6 uni
Router(config)#ipv6 unicast-routing
Router(config)#interface
Router(config)#interface gi
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#ipv6 ad
Router(config-if)#ipv6 address
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ipv6 address
Router(config-if)#ipv6 address 2001:DB8:A::1/64
Router(config-if)#NO SH
Router(config-if)#NO SHutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed
state to up

Router(config-if)#
Router(config-if)#interface GigabitEthernet0/0/1
Router(config-if)#ipv6 address 2001:DB8:B::1/64
Router(config-if)#NO SHutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed
state to up
```

e. Testing IPv6 Connectivity using PC2

PC2

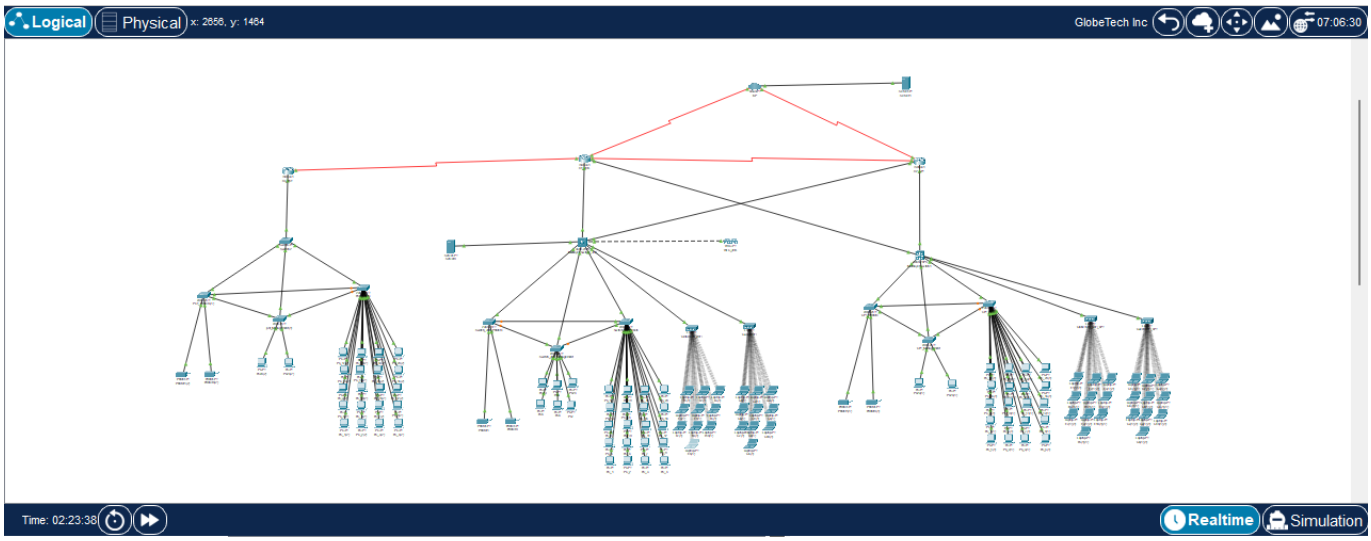
Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 2001:DB8:A::11 with 32 bytes of data:

Reply from 2001:DB8:A::11: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:A::11: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:A::11: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:A::11: bytes=32 time=1ms TTL=127

Ping statistics for 2001:DB8:A::11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:A::10

Pinging 2001:DB8:A::10 with 32 bytes of data:

Reply from 2001:DB8:A::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:A::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:A::10: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:A::10: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:A::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:B::11

Pinging 2001:DB8:B::11 with 32 bytes of data:

Reply from 2001:DB8:B::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:B::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:B::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:B::11: bytes=32 time=1ms TTL=128

Ping statistics for 2001:DB8:B::11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Network Topology and Design (IPv4)

a) Below you will find the topology for GlobeTech's network, this topology was designed to provide redundancy in the network and avoid single points of failure



b) . Given the address 172.168.0.0/16 I've used VLSM (Variable Length Subnet Mask) which is a subnet design strategy where all subnet masks can have varying sizes as defined by (Awati, 2021)

VLAN	Site/ Purpose	Number of Hosts	Needed bits	Network Address	CIDR	Subnet Mask	Address Range	Broadcast Address
10	GlobeTech -WIFI	100	9	172.16.2.0	/25	255.255.255.128	172.16.2.1 – 172.16.2.126	172.16.2.127
20	Johannesburg	200	7	172.16.0.0	/23	255.255.254.0	172.16.0.1- 172.16.1.254	172.16.1.255
30	Cape Town	40	10	172.16.3.0	/26	255.255.255.192	172.16.3.1 – 172.16.3.62	172.16.3.63
40	Potchefstroom	20	11	172.16.3.64	/27	255.255.255.224	172.16.3.65 – 172.16.3.94	172.3.16.3.95
60	Printers	20	11	172.16.3.96	/27	255.255.255.224	172.16.3.97 - 172.16.3.126	172.16.3.127
100	Guest-WIFI	100	9	172.16.2.128	/25	255.255.255.128	172.16.2.129 –	172.16.2.255

							172.16.2.254	
99	Management	10	12	172.16.3.128	/28	255.255.255.240	172.16.3.129 – 192.16.3.142	172.16.3.143

The table below contains Core Routers and Distribution Layer IP addresses which were created using VLSM from 10.0.0.0/8

Link / Site Purpose	Hosts	Needed Bits	Network Address	CIDR	Subnet Mask	Address Range	Broadcast Address
Core R1_JHB - Core R2_CPT	2	2	10.0.0.0	/30	255.255.255.252	10.0.0.1 – 10.0.0.2	10.0.0.3
R1_JHB-ISP	2	2	10.0.0.4	/30	255.255.255.252	10.0.0.5 – 10.0.0.6	10.0.0.7
R2_CPT-ISP	2	2	10.0.0.8	/30	255.255.255.252	10.0.0.9 – 10.0.0.10	10.0.0.11
R1_JHB-R3_POT	2	2	10.0.0.12	/30	255.255.255.252	10.0.0.13 – 10.0.0.14	10.0.0.15
R1_JHB↔-MS1_JHB (L3 Switch)	2	2	10.0.0.16	/30	255.255.255.252	10.0.0.17 – 10.0.0.18	10.0.0.19
R2_CPT ↔ MS1_JHB (L3 Switch)	2	2	10.0.0.20	/30	255.255.255.252	10.0.0.21 – 10.0.0.22	10.0.0.23
R1_JHB ↔ MS2_CPT (L3 Switch)	2	2	10.0.0.24	/30	255.255.255.252	10.0.0.25 – 10.0.0.26	10.0.0.27
R2_CPT ↔ MS2_CPT (L3 Switch)	2	2	10.0.0.28	/30	255.255.255.252	10.0.0.29 – 10.0.0.30	10.0.0.31

Loopback – R1_JHB	1	0	10.255.0.1	/32	255.255.255.255	10.255.0.1	—
Loopback – R2_CPT	1	0	10.255.0.5	/32	255.255.255.255	10.255.0.5	—
Loopback – R3_POT	1	0	10.255.0.3	/32	255.255.255.255	10.255.0.3	—
Loopback – MS1_JHB	1	0	10.255.0.11	/32	255.255.255.255	10.255.0.11	—
Loopback – MS2_CPT	1	0	10.255.0.12	/32	255.255.255.255	10.255.0.12	—

Purpose	Site	Subnet	CIDR	Address Range	Broadcast
GlobeTech-WIFI	JHB	172.16.2.0	/25	172.16.2.1 – 126	172.16.2.127
	CPT	172.16.4.0	/25	172.16.4.1 – 126	172.16.4.127
Printers	JHB	172.16.3.96	/27	172.16.3.97 – 126	172.16.3.127
	CPT	172.16.5.96	/27	172.16.5.97 – 126	172.16.5.127
	POT	172.16.6.96	/27	172.16.6.97 – 126	172.16.6.127
Management	JHB	172.16.3.128	/28	172.16.3.129 – 142	172.16.3.143
	CPT	172.16.5.128	/28	172.16.5.129 – 142	172.16.5.143
	POT	172.16.6.128	/28	172.16.6.129 – 142	172.16.6.143
Guest WiFi	JHB	172.16.2.128	/25	172.16.2.129 – 254	172.16.2.255

	CPT	172.16.7.128	/25	172.16.7.129 – 254	172.16.7.255

The below table holds subnets for the following VLAMs 10, 60,99,100 for different location

Device configuration

Basic Settings: The fundamental device settings have been configured on all interconnectivity devices with the help of (CISCO, 2015), the configuration includes hostnames, enabled passwords and username and password using “cisco”. Below you’ll find examples of the configuration using the Johannesburg interconnectivity devices, please note that the configurations have been applied to all necessary devices within packet tracer.

R1_JHB

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up


Router>en
Router>enable
Router#conf
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ho
Router(config)#hostname R1_JHB
R1_JHB(config)#en
R1_JHB(config)#ena
R1_JHB(config)#enable secret cisco
R1_JHB(config)#username cisco password cisco
R1_JHB(config)#line con
R1_JHB(config)#line console 0
R1_JHB(config-line)#login local
R1_JHB(config-line)#exit
R1_JHB(config)#exit
R1_JHB#
%SYS-5-CONFIG_I: Configured from console by console

R1_JHB#write m
R1_JHB#write memory
Building configuration...
[OK]
R1_JHB#
```

Copy Paste

Top

```
R1_JHB(config)#no ip domain-lookup
R1_JHB(config)#exit
R1_JHB#
%SYS-5-CONFIG_I: Configured from console by console

R1_JHB#writ
R1_JHB#write me
R1_JHB#write memory
Building configuration...
[OK]
```

Multilayer Switch_JHB

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet1/0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to up


Switch>en
Switch>enable
Switch#config
Switch#configure te
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hos
Switch(config)#hostname MS1_JHB
MS1_JHB(config)#en
MS1_JHB(config)#ena
MS1_JHB(config)#enable se
MS1_JHB(config)#enable secret cisco
MS1_JHB(config)#username cisco pass
MS1_JHB(config)#username cisco password cisco
MS1_JHB(config)#line console 0
MS1_JHB(config-line)#login local
MS1_JHB(config-line)#exit
MS1_JHB(config)#no ip domain
MS1_JHB(config)#no ip domain-lookup
MS1_JHB(config)#exit
MS1_JHB#
%SYS-5-CONFIG_I: Configured from console by console

MS1_JHB#wr
MS1_JHB#write m
MS1_JHB#write memory
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
MS1_JHB#
```

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Switch_JHB_Printers

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


Switch>en
Switch>enable
Switch#confi
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ho
Switch(config)#hostname S_JHB_Printers
S_JHB_Printers(config)#ena
S_JHB_Printers(config)#enable s
S_JHB_Printers(config)#enable secret cisco
S_JHB_Printers(config)#username cisco pas
S_JHB_Printers(config)#username cisco password cisco
S_JHB_Printers(config)#no ip domain-lookup
S_JHB_Printers(config)#line console 0
S_JHB_Printers(config-line)#login local
S_JHB_Printers(config-line)#exit
S_JHB_Printers(config)#exit
S_JHB_Printers#
%SYS-5-CONFIG_I: Configured from console by console

S_JHB_Printers#wri
S_JHB_Printers#write m
S_JHB_Printers#write memory
Building configuration...
[OK]
S_JHB_Printers#
```

Switch_JHB_Management

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up


Switch>en
Switch>enable
Switch#config
Switch#configure t
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ho
Switch(config)#hostname S_JHB_Management
S_JHB_Management(config)#ena
S_JHB_Management(config)#enable s
S_JHB_Management(config)#enable secret cisco
S_JHB_Management(config)#username cisco password cisco
S_JHB_Management(config)#no ip domain-lookup
S_JHB_Management(config)#line console 0
S_JHB_Management(config-line)#login local
S_JHB_Management(config-line)#exit
S_JHB_Management(config)#exit
S_JHB_Management#
%SYS-5-CONFIG_I: Configured from console by console

S_JHB_Management#write memory
Building configuration...
[OK]
S_JHB_Management#
```

Switch_JHB_Hosts

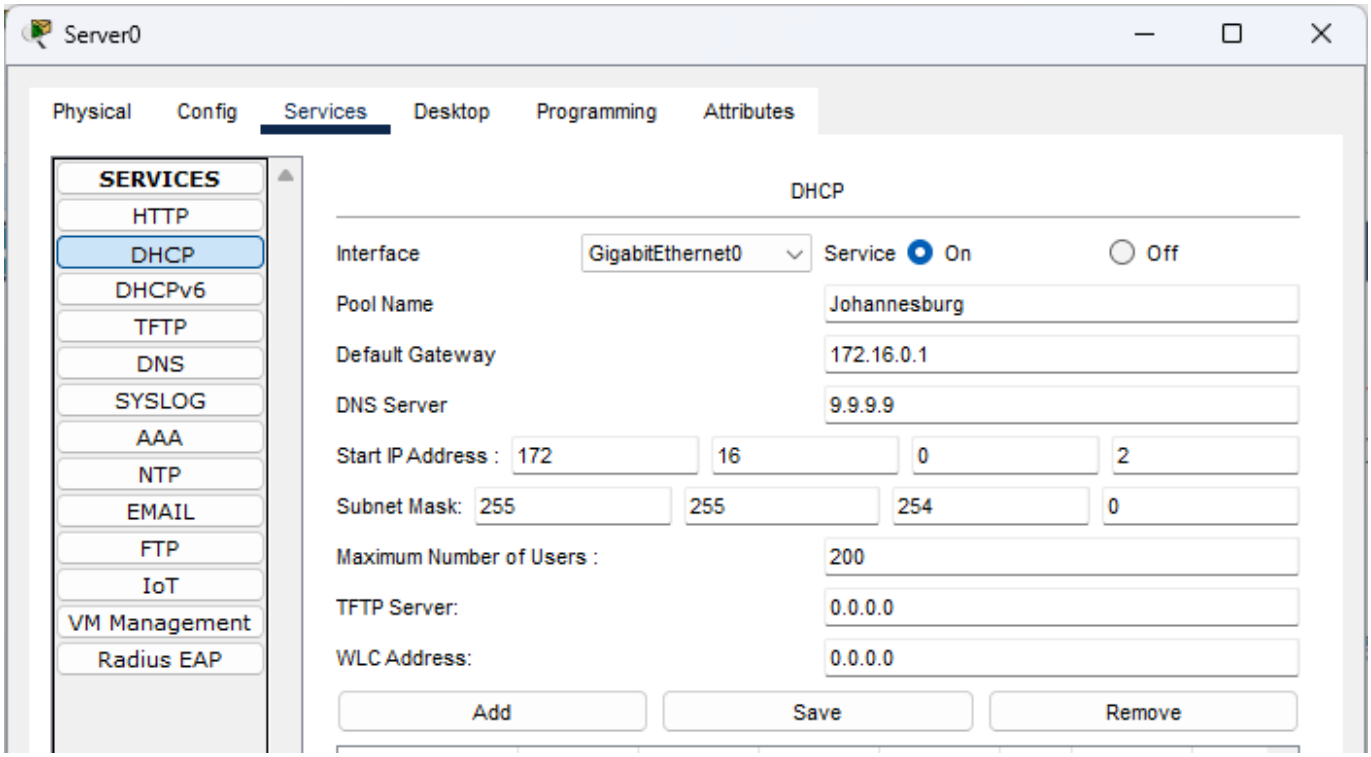
Physical Config CLI Attributes

IOS Command Line Interface

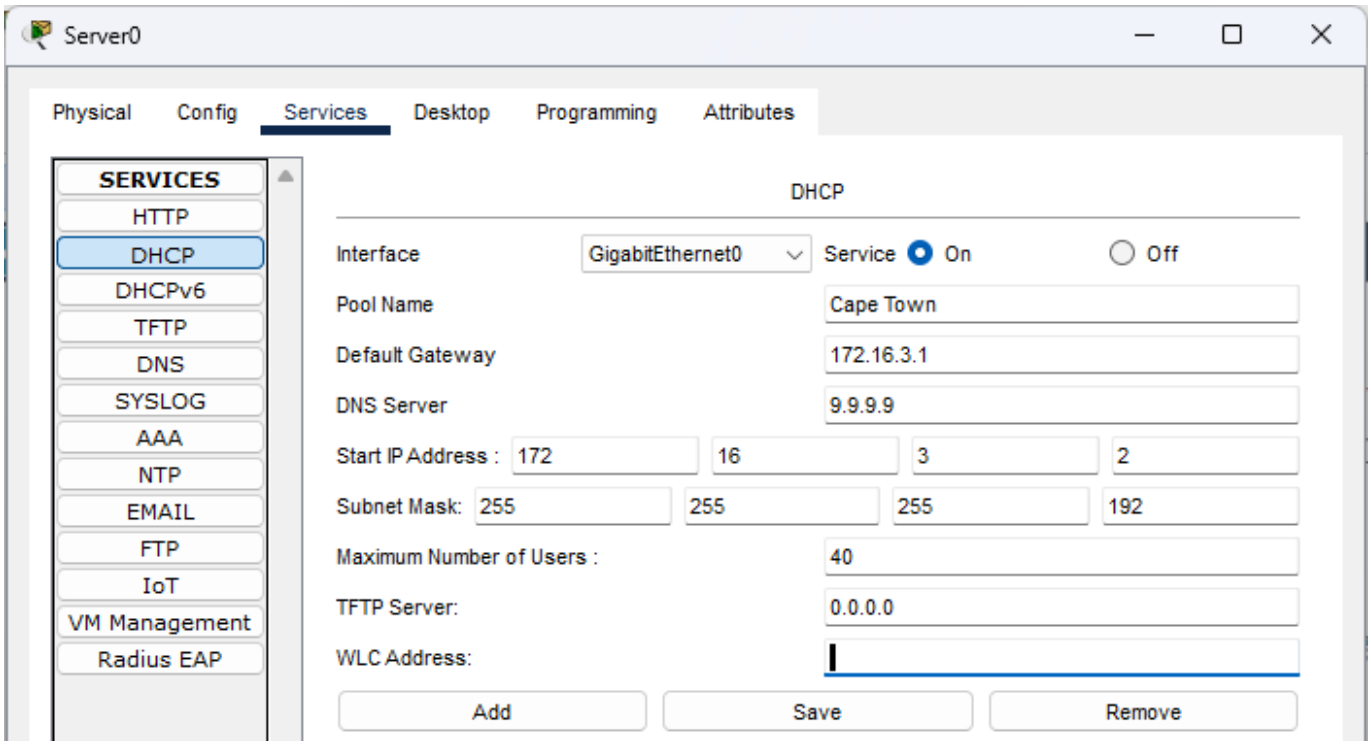
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


Switch>ena
Switch>enable
Switch#conf
Switch#configure t
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ho
Switch(config)#hostname S_JHB_Hosts
S_JHB_Hosts(config)#ena
S_JHB_Hosts(config)#enable s
S_JHB_Hosts(config)#enable secret cisco
S_JHB_Hosts(config)#username cisco password cisco
S_JHB_Hosts(config)#no ip domain-lookup
S_JHB_Hosts(config)#line console 0
S_JHB_Hosts(config-line)#login local
S_JHB_Hosts(config-line)#exit
S_JHB_Hosts(config)#exit
S_JHB_Hosts#
%SYS-5-CONFIG_I: Configured from console by console

S_JHB_Hosts#wr
S_JHB_Hosts#write m
S_JHB_Hosts#write memory
Building configuration...
[OK]
S_JHB_Hosts#
```

DHCP configuration**:** DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to dynamically assign an internet Protocol (IP) address to any device on a network so it can communicate, as defined by (Gilies, 2025). The DHCP server has been configured to allocate IP addresses to all the end devices across the entire organization with the help of (Kimanzi, 2018). Below you will find the configuration:

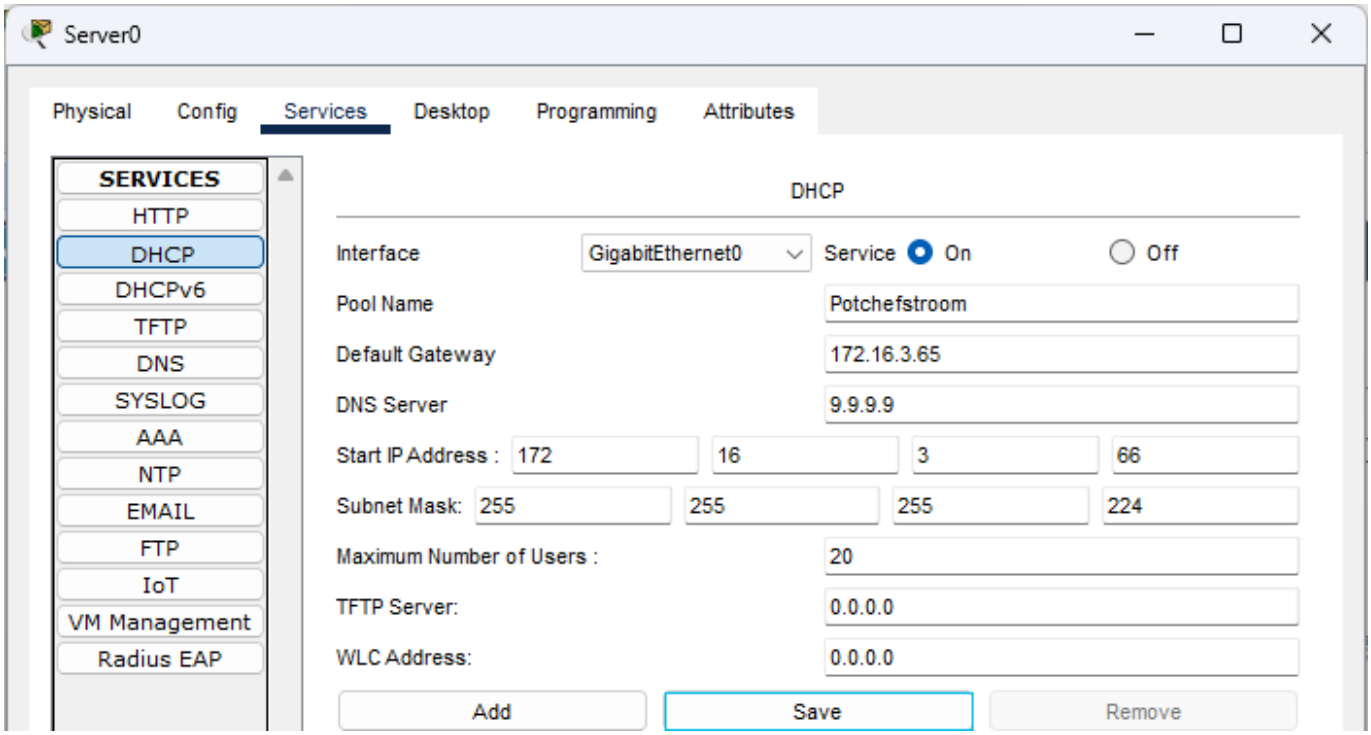
Johannesburg:


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
GigabitEthernet0
Service
On
Off
Pool Name
Johannesburg
Default Gateway
172.16.0.1
DNS Server
9.9.9.9
Start IP Address :
172
16
0
2
Subnet Mask:
255
255
254
0
Maximum Number of Users :
200
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove

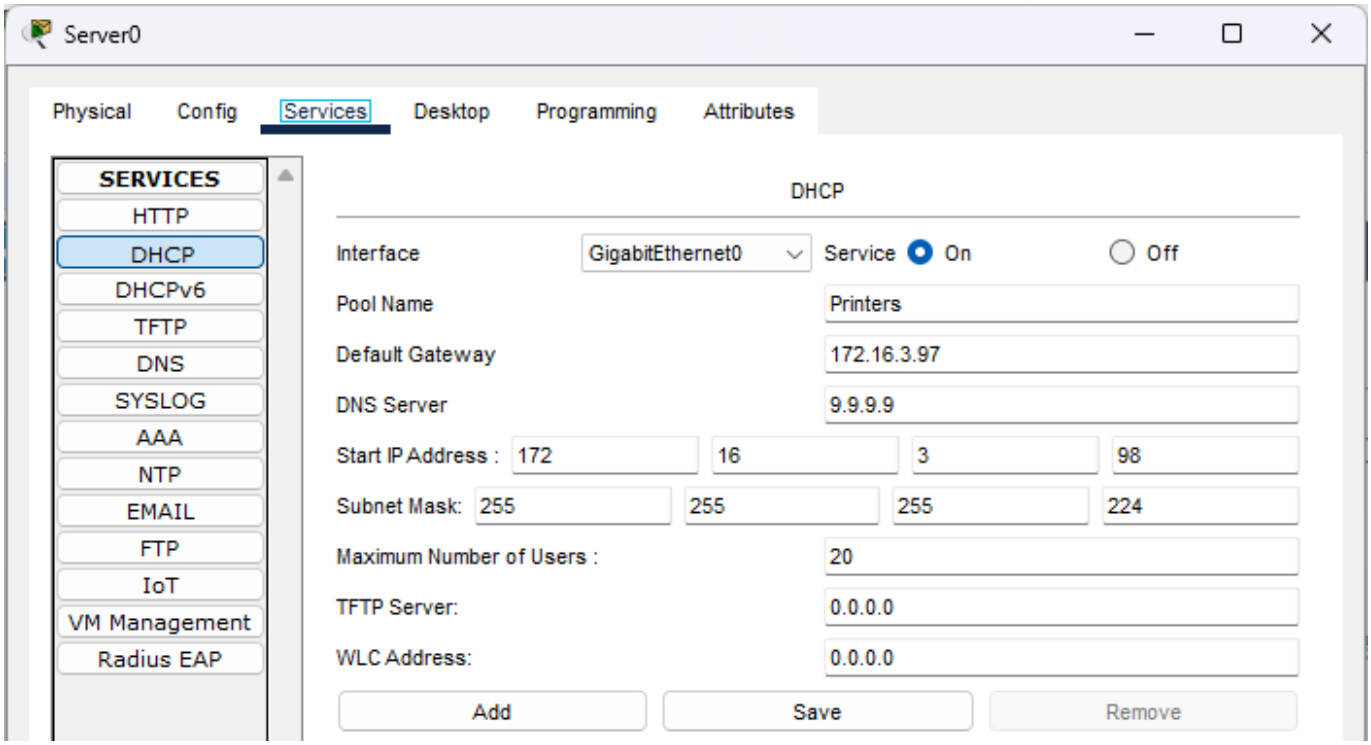
Cape Town:


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
GigabitEthernet0
Service
On
Off
Pool Name
Cape Town
Default Gateway
172.16.3.1
DNS Server
9.9.9.9
Start IP Address :
172
16
3
2
Subnet Mask:
255
255
255
192
Maximum Number of Users :
40
TFTP Server:
0.0.0.0
WLC Address:
Add
Save
Remove

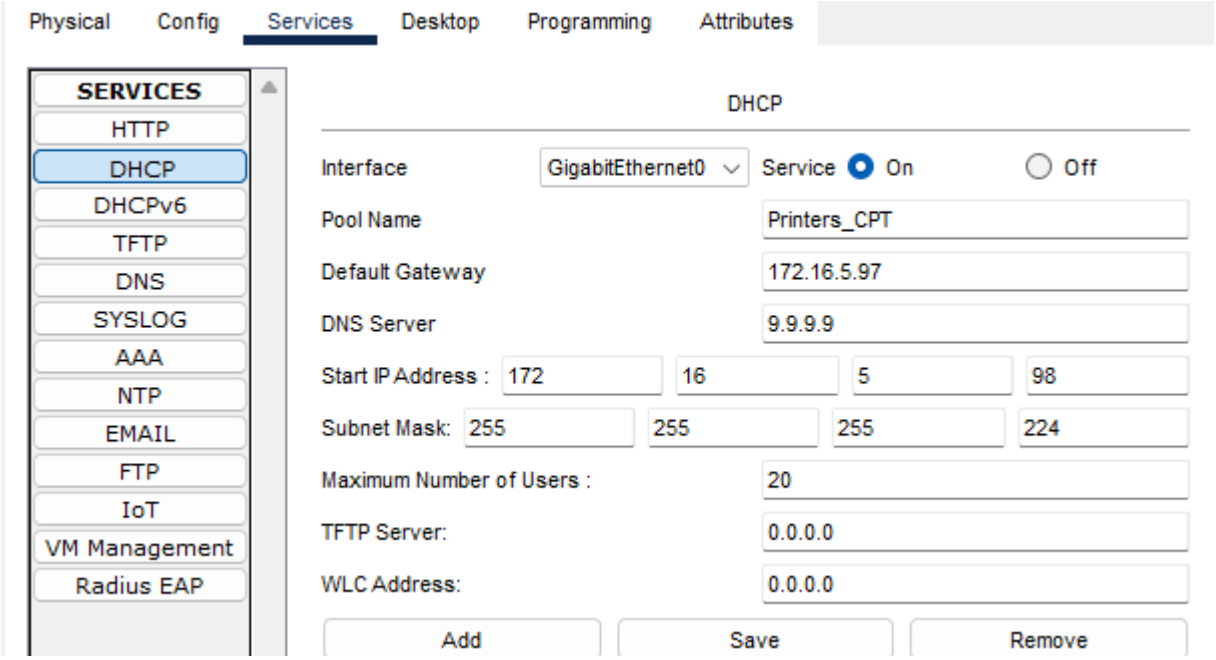
Potchefstroom:



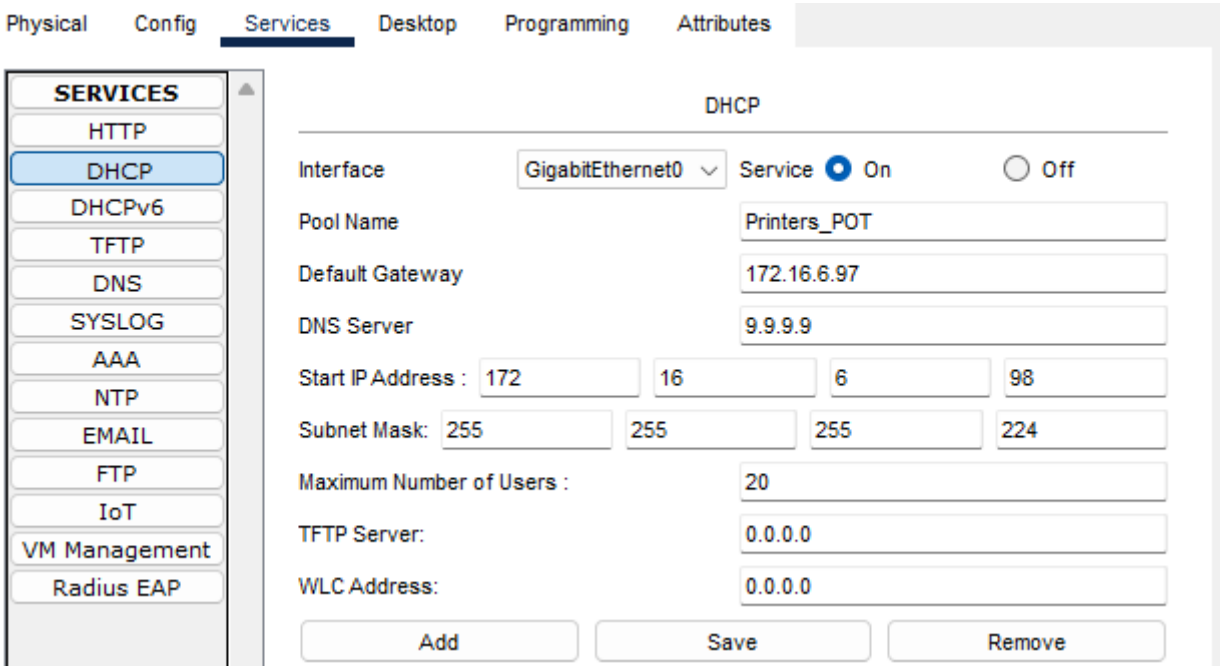
Printers:



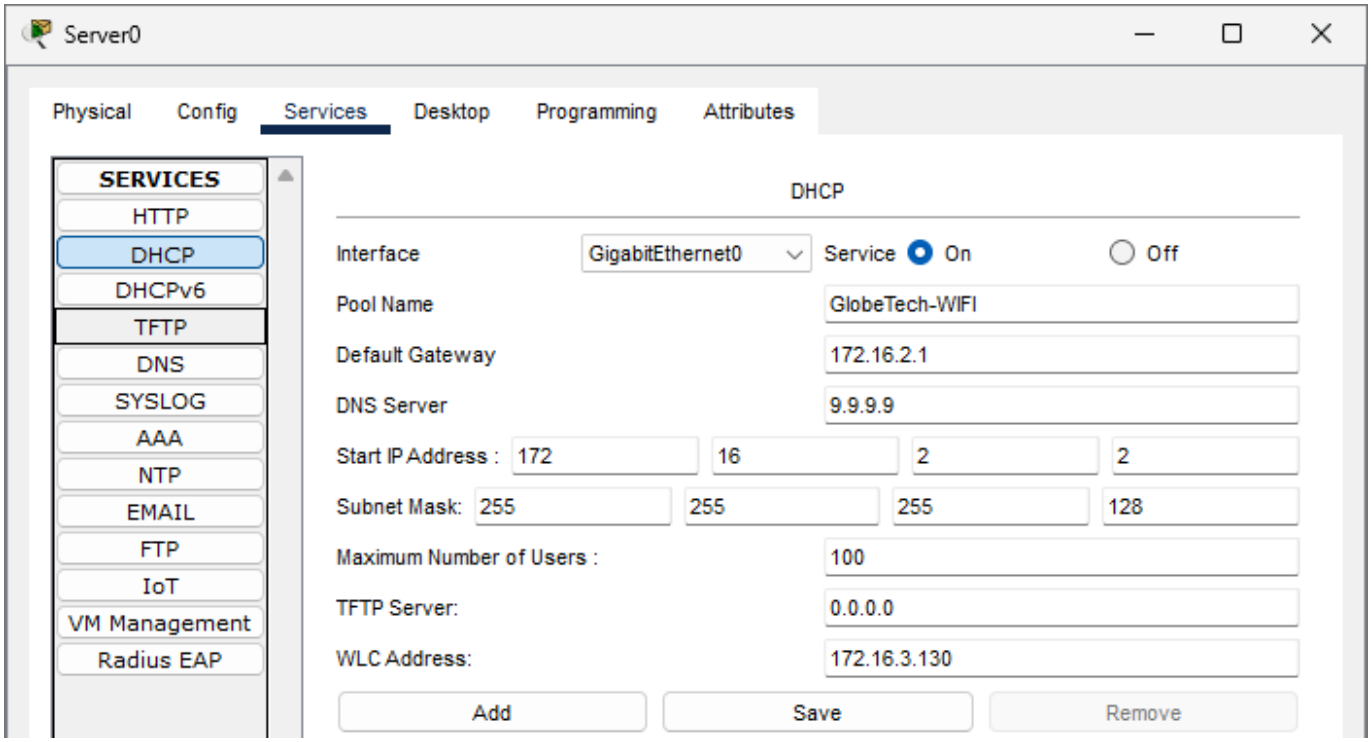
Printers_CPT:



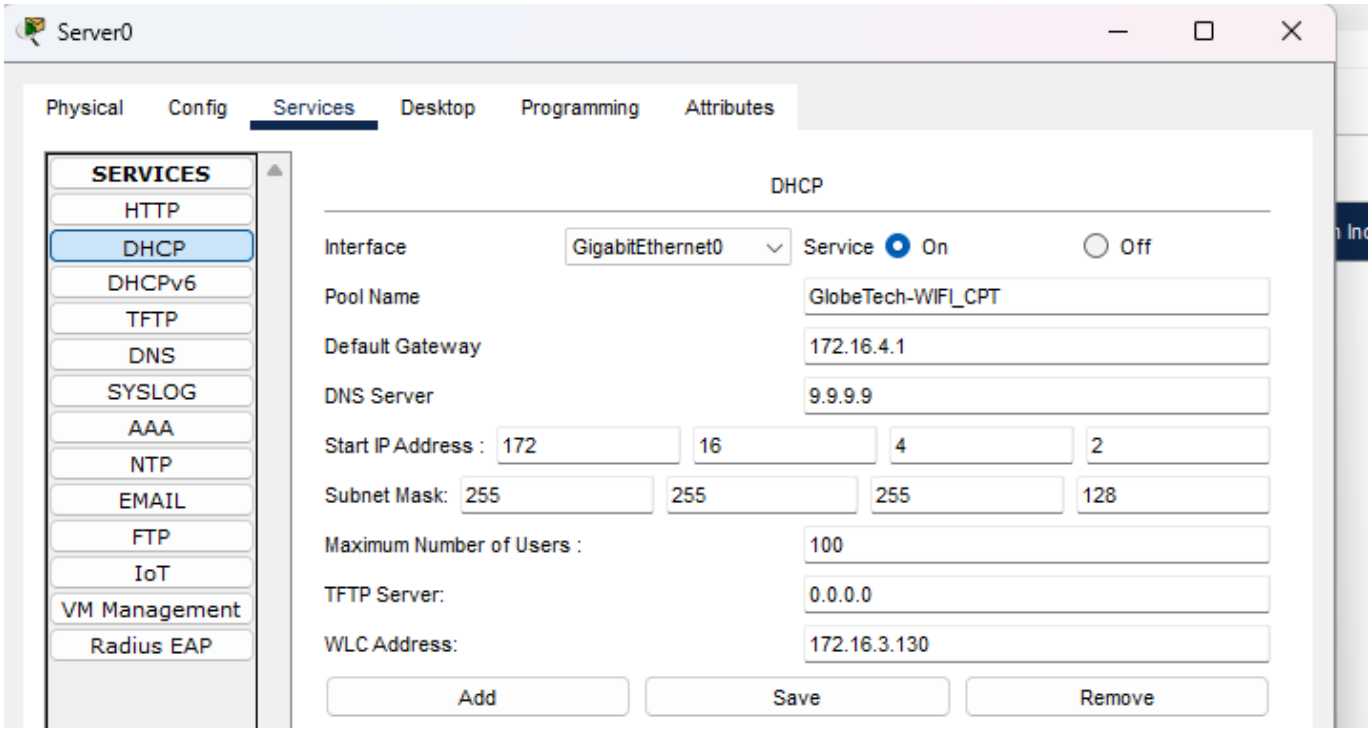
Printers_POT:



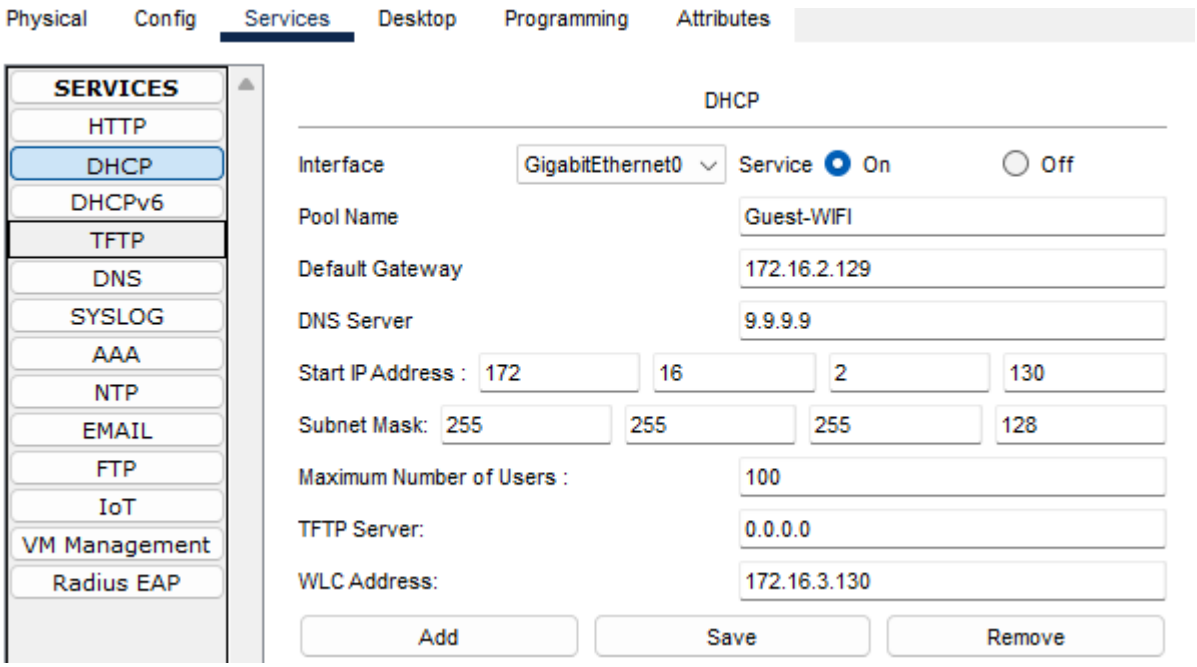
GbobeTech-WIFI:



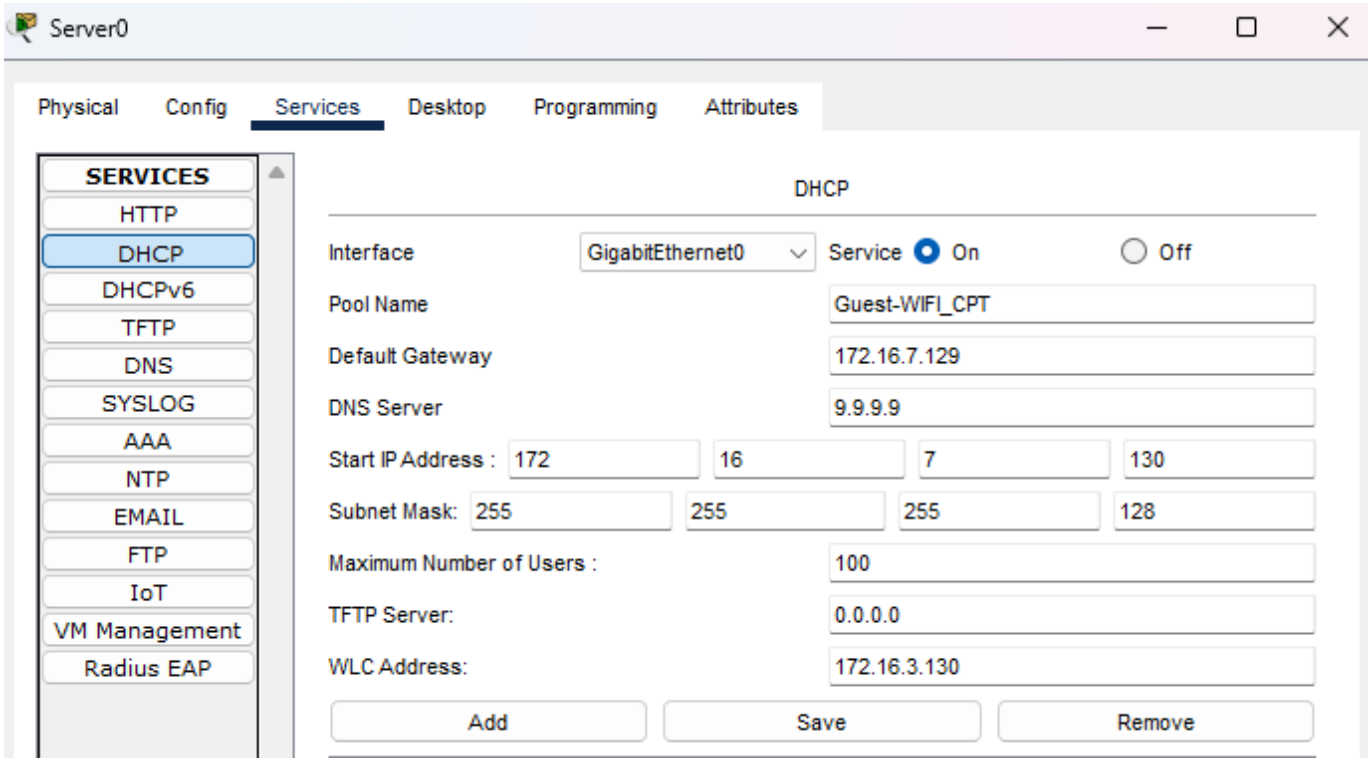
GbobeTech-WIFI_CPT:



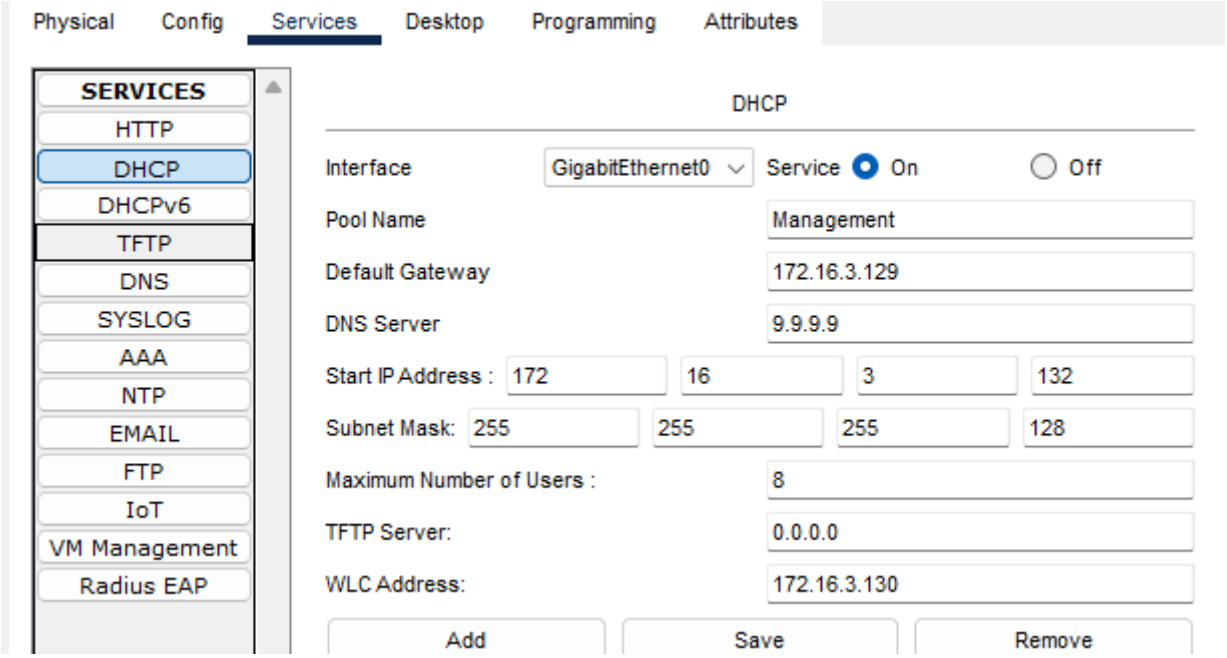
Guest-WIFI:



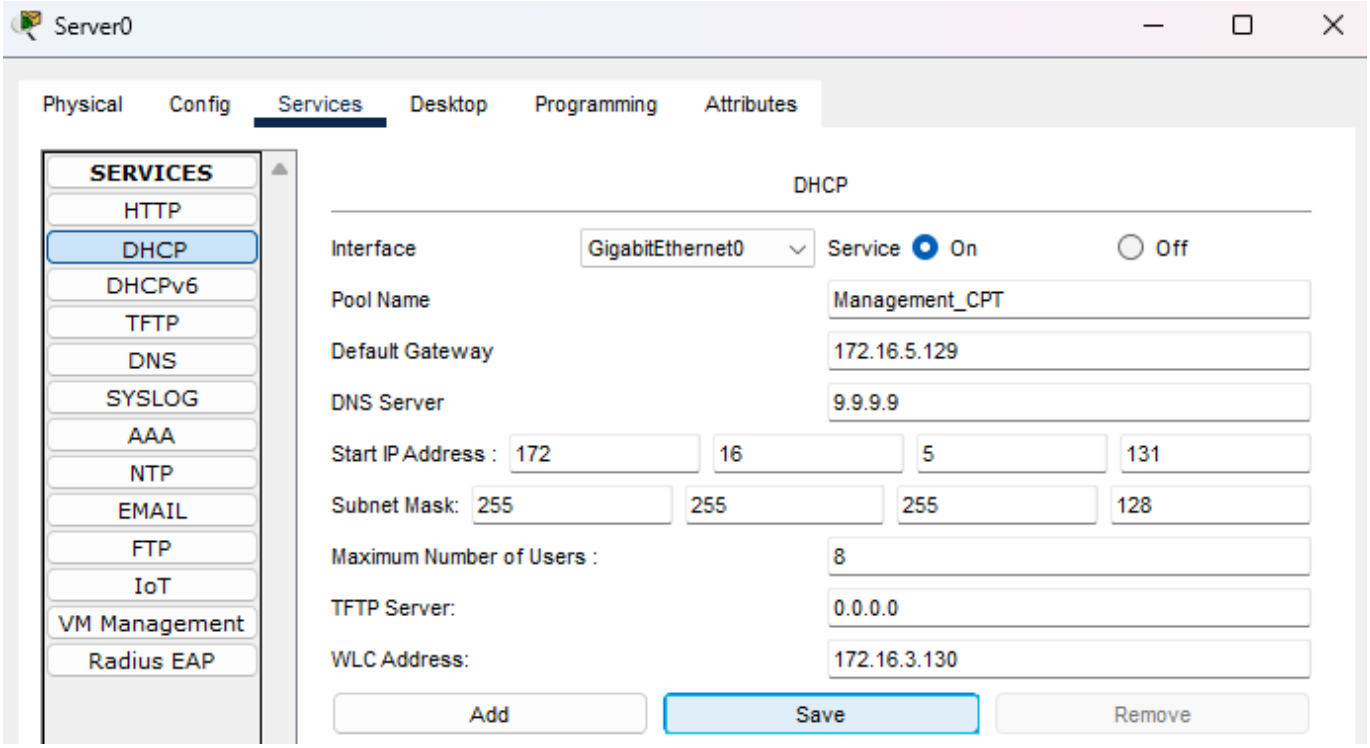
Guest-WIFI_CPT:



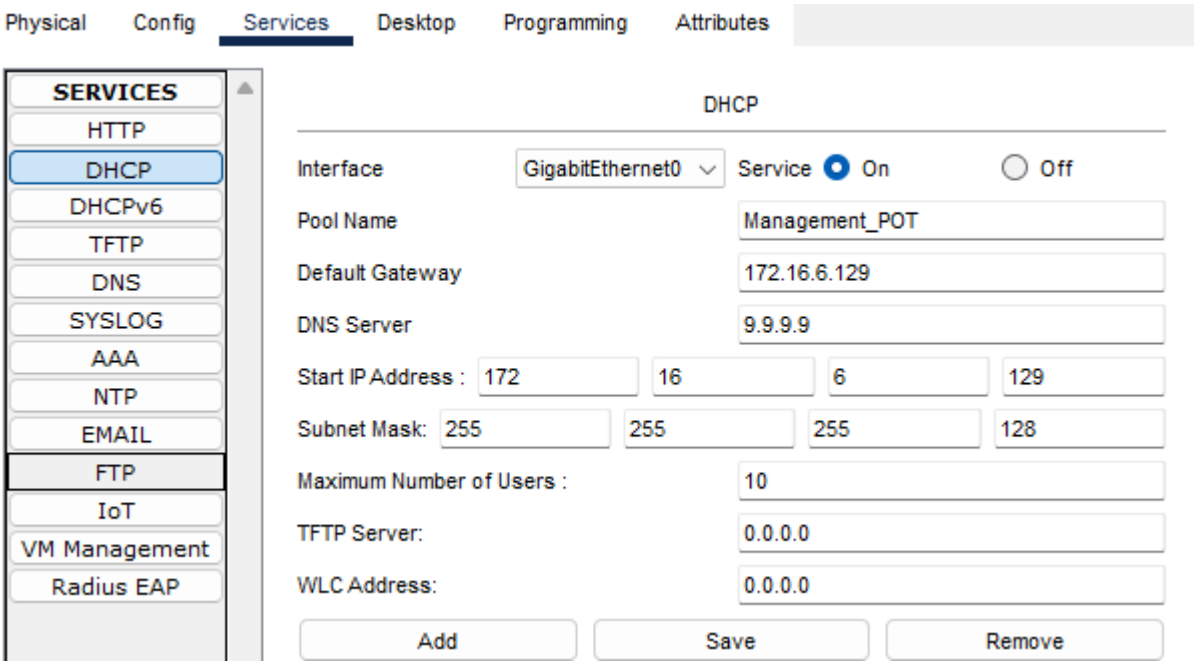
Management:



Management_CPT:



Management_POT:



VLAN (Virtual Local Area Network):is a technique used to create separate logical networks in a single physical network infrastructure. Below you'll find VLANs that were created and named as specified in the scenario configured with the help from (Anshika, 2024). The images provided only show 1 switch configuration per site, other configurations are available on the packet tracer file.

VLANS at Johannesburg:

```
S_JHB_Printers(config)#vlan 10
S_JHB_Printers(config-vlan)#name GlobeTech-WIFI
S_JHB_Printers(config-vlan)#S_JHB_Printers(config-vlan)#
S_JHB_Printers(config-vlan)#end
S_JHB_Printers#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S_JHB_Printers(config)#
S_JHB_Printers(config)#
%SYS-5-CONFIG_I: Configured from console by console

S_JHB_Printers(config)#vlan 20
S_JHB_Printers(config-vlan)#name Johannesburg
S_JHB_Printers(config-vlan)#exit
S_JHB_Printers(config)#vlan 60
S_JHB_Printers(config-vlan)#name Printers
S_JHB_Printers(config-vlan)#exit
S_JHB_Printers(config)#vlan 100
S_JHB_Printers(config-vlan)#name Guest-WIFI
S_JHB_Printers(config-vlan)#exit
S_JHB_Printers(config)#vlan 99
S_JHB_Printers(config-vlan)#name Management
S_JHB_Printers(config-vlan)#exit
S_JHB_Printers(config)#exit
S_JHB_Printers#
```

VLANS at Cape Town:

CPT_Printers

Physical Config CLI Attributes

IOS Command Line Interface

```
S_CPT_Printers(config)#vlan 10
S_CPT_Printers(config-vlan)#name GlobeTech-W
S_CPT_Printers(config-vlan)#name GlobeTech-WIFI
S_CPT_Printers(config-vlan)#vlan 30
S_CPT_Printers(config-vlan)#name Cape_Town
S_CPT_Printers(config-vlan)#vlan 60
S_CPT_Printers(config-vlan)#name Printers
S_CPT_Printers(config-vlan)#vlan 99
S_CPT_Printers(config-vlan)#name Management
S_CPT_Printers(config-vlan)#vlan 100
S_CPT_Printers(config-vlan)#name Guest-WIFI
S_CPT_Printers(config-vlan)#exit
S_CPT_Printers(config)#exit
S_CPT_Printers#
%SYS-5-CONFIG_I: Configured from console by console

S_CPT_Printers#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   GlobeTech-WIFI                   active
30   Cape_Town                        active
60   Printers                         active
99   Management                       active
100  Guest-WIFI                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S_CPT_Printers#
```

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VLANS at Potchefstroom:

POT_Printers(1)

Physical Config CLI Attributes

IOS Command Line Interface

```
Enter configuration commands, one per line.  End with CNTL/Z.
S_POT_Printers(config)#
S_POT_Printers(config)#vlan 10
S_POT_Printers(config-vlan)# name GloveTech-WIFI
S_POT_Printers(config-vlan)#vlan 40
S_POT_Printers(config-vlan)# name Potchestroom
S_POT_Printers(config-vlan)#vlan 60
S_POT_Printers(config-vlan)# name Printers
S_POT_Printers(config-vlan)#vlan 100
S_POT_Printers(config-vlan)# name Guest-WIFI
S_POT_Printers(config-vlan)#vlan 99
S_POT_Printers(config-vlan)# name Management
S_POT_Printers(config-vlan)#exit
S_POT_Printers(config)#exit
S_POT_Printers#
%SYS-5-CONFIG_I: Configured from console by console
show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   GloveTech-WIFI                   active
40   Potchestroom                     active
60   Printers                         active
99   Management                       active
100  Guest-WIFI                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S_POT_Printers#
```

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Switch ports :are layer 2 interfaces used to move layer 2 traffic. The are varies types of switch ports which include access ports, trunk ports and dynamic ports. Access ports can only carry traffic from one VLAN which they have been assigned to, they connect end-user devices to the switch. While trunk ports can carry traffic from various VLANS, they connect switches to switches/ routers (GeeksforGeeks, 2023). Below you'll find assigned access and trunk ports on all the necessary network devise:

For Johannesburg network devices:

Multilayer Switch_JHB

Physical Config CLI Attributes

IOS Command Line Interface

```
MS1_JHB#show interface brief
                         ^
% Invalid input detected at '^' marker.

MS1_JHB#show interface st
MS1_JHB#show interface status
Port      Name               Status       Vlan       Duplex  Speed Type
Gig1/0/1                     connected    1          auto    auto  10/100BaseTX
Gig1/0/2                     connected    1          auto    auto  10/100BaseTX
Gig1/0/3                     connected    99         auto    auto  10/100BaseTX
Gig1/0/4                     connected    99         auto    auto  10/100BaseTX
Gig1/0/5                     connected    trunk      auto    auto  10/100BaseTX
Gig1/0/6                     connected    trunk      auto    auto  10/100BaseTX
Gig1/0/7                     connected    trunk      auto    auto  10/100BaseTX
Gig1/0/8                     connected    trunk      auto    auto  10/100BaseTX
Gig1/0/9                     connected    trunk      auto    auto  10/100BaseTX
Gig1/0/10                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/11                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/12                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/13                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/14                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/15                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/16                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/17                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/18                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/19                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/20                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/21                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/22                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/23                    notconnect   1          auto    auto  10/100BaseTX
Gig1/0/24                    notconnect   1          auto    auto  10/100BaseTX
Gig1/1/1                     notconnect   1          auto    auto  10/100BaseTX
Gig1/1/2                     notconnect   1          auto    auto  10/100BaseTX
Gig1/1/3                     notconnect   1          auto    auto  10/100BaseTX
Gig1/1/4                     notconnect   1          auto    auto  10/100BaseTX

MS1_JHB#
```

Switch_JHB_Management

Physical Config CLI Attributes

IOS Command Line Interface

```
S_JHB_Management#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S_JHB_Management(config)#interface ra
S_JHB_Management(config)#interface range f
S_JHB_Management(config)#interface range fastEthernet0/1 - 23
S_JHB_Management(config-if-range)#switchport mode access
S_JHB_Management(config-if-range)#switchport access vlan 99
S_JHB_Management(config-if-range)#exit
S_JHB_Management(config)#interface rage gi
S_JHB_Management(config)#interface range gi
S_JHB_Management(config)#interface range gigabitEthernet0/1, gi
S_JHB_Management(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2, f
S_JHB_Management(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2,
fastEthernet0/24
S_JHB_Management(config-if-range)#switchport
% Incomplete command.
S_JHB_Management(config-if-range)#switchport mode trunk


S_JHB_Management(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

S_JHB_Management(config-if-range)#switchport trunk allowed vlan 99,20,60
S_JHB_Management(config-if-range)#exit
S_JHB_Management(config)#exit
```

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Switch_JHB_Printers

Physical Config CLI Attributes

IOS Command Line Interface

```
S_JHB_Printers>en
Password:
S_JHB_Printers#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S_JHB_Printers(config)#interface range fast
S_JHB_Printers(config)#interface range fastEthernet0/1-23
S_JHB_Printers(config-if-range)#switchport mode access
S_JHB_Printers(config-if-range)#switchport access vlan 60
S_JHB_Printers(config-if-range)#exit
S_JHB_Printers(config)#interface range gi
S_JHB_Printers(config)#interface range gigabitEthernet0/1, gi
S_JHB_Printers(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2, f
S_JHB_Printers(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2,
fastEthernet0/24
S_JHB_Printers(config-if-range)#switchport mode trunk


S_JHB_Printers(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

S_JHB_Printers(config-if-range)#switchport trunk allowed vlan 60,20,99
S_JHB_Printers(config-if-range)#exit
S_JHB_Printers(config)#exit
S_JHB_Printers#
%SYS-5-CONFIG_I: Configured from console by console

S_JHB_Printers#show inter
S_JHB_Printers#show interfaces st
S_JHB_Printers#show interfaces status
Port      Name                Status       Vlan       Duplex  Speed Type
Fa0/1                         connected    60         auto    auto  10/100BaseTX
```

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Switch_JHB_Hosts

Physical Config CLI Attributes

IOS Command Line Interface

```
S_JHB_Hosts>en
Password:
S_JHB_Hosts#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S_JHB_Hosts(config)#inter
S_JHB_Hosts(config)#interface range f
S_JHB_Hosts(config)#interface range fastEthernet0/1 - 23
S_JHB_Hosts(config-if-range)#switchport mode access
S_JHB_Hosts(config-if-range)#switchport access vlan 20
S_JHB_Hosts(config-if-range)#exit
S_JHB_Hosts(config)#interf
S_JHB_Hosts(config)#interface range gi
S_JHB_Hosts(config)#interface range gigabitEthernet0/1 - 2, fa
S_JHB_Hosts(config)#interface range gigabitEthernet0/1 - 2, fastEthernet0/24
S_JHB_Hosts(config-if-range)#switchport mode trunk

S_JHB_Hosts(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

S_JHB_Hosts(config-if-range)#exit
S_JHB_Hosts(config)#interface range gigabitEthernet0/1,go fastEthernet0/24
S_JHB_Hosts(config)#interface range gigabitEthernet0/1,gi fastEthernet0/24
S_JHB_Hosts(config)#interface range gigabitEthernet0/1,gi
S_JHB_Hosts(config)#interface range gigabitEthernet0/1, gig
S_JHB_Hosts(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2, f
S_JHB_Hosts(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2,
fastEthernet0/24
S_JHB_Hosts(config-if-range)#switchport mode trunk
S_JHB_Hosts(config-if-range)#switchport trunk allowed vlan 20,60,99
S_JHB_Hosts(config-if-range)#exit
S_JHB_Hosts(config)#exit
S_JHB_Hosts#
%SYS-5-CONFIG_I: Configured from console by console
```

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For Cape Town network devices:

Multilayer Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
1002 fddi-default                       active
1003 token-ring-default                 active
1004 fddinet-default                    active
1005 trnet-default                      active
MS2_CPT#show interface sta
MS2_CPT#show interface status
Port      Name                 Status       Vlan       Duplex  Speed Type
Gig1/0/1                       connected    routed     auto    auto  10/100BaseTX
Gig1/0/2                       connected    routed     auto    auto  10/100BaseTX
Gig1/0/3                       connected    trunk      auto    auto  10/100BaseTX
Gig1/0/4                       connected    trunk      auto    auto  10/100BaseTX
Gig1/0/5                       connected    trunk      auto    auto  10/100BaseTX
Gig1/0/6                       connected    trunk      auto    auto  10/100BaseTX
Gig1/0/7                       connected    99         auto    auto  10/100BaseTX
Gig1/0/8                       connected    trunk      auto    auto  10/100BaseTX
Gig1/0/9                       notconnect   1          auto    auto  10/100BaseTX
Gig1/0/10                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/11                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/12                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/13                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/14                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/15                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/16                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/17                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/18                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/19                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/20                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/21                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/22                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/23                      notconnect   1          auto    auto  10/100BaseTX
Gig1/0/24                      notconnect   1          auto    auto  10/100BaseTX
Gig1/1/1                       notconnect   1          auto    auto  10/100BaseTX
Gig1/1/2                       notconnect   1          auto    auto  10/100BaseTX
Gig1/1/3                       notconnect   1          auto    auto  10/100BaseTX
Gig1/1/4                       notconnect   1          auto    auto  10/100BaseTX

MS2_CPT#
```

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CPT_HOST

Physical Config CLI Attributes

IOS Command Line Interface

```
S_CPT_Hosts>en
S_CPT_Hosts>enable
Password:
S_CPT_Hosts#in
S_CPT_Hosts#int
S_CPT_Hosts#interf
S_CPT_Hosts#confi
S_CPT_Hosts#configure t
S_CPT_Hosts#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S_CPT_Hosts(config)#inter
S_CPT_Hosts(config)#interface r
S_CPT_Hosts(config)#interface range f
S_CPT_Hosts(config)#interface range fastEthernet0/1 - 23
S_CPT_Hosts(config-if-range)#switch
S_CPT_Hosts(config-if-range)#switchport mod
S_CPT_Hosts(config-if-range)#switchport mode access
S_CPT_Hosts(config-if-range)#switchport mode access
S_CPT_Hosts(config-if-range)#s
S_CPT_Hosts(config-if-range)#switchport access vlan 30
S_CPT_Hosts(config-if-range)#exit
S_CPT_Hosts(config)#inter
S_CPT_Hosts(config)#interface r
S_CPT_Hosts(config)#interface range gi
S_CPT_Hosts(config)#interface range gigabitEthernet0/1, gi
S_CPT_Hosts(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2, f
S_CPT_Hosts(config)#interface range gigabitEthernet0/1, gigabitEthernet0/2,
fastEthernet0/24
S_CPT_Hosts(config-if-range)#switchport mode trunk



S_CPT_Hosts(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
```

Copy Paste

Top

CPT_Printers

Physical Config CLI Attributes

IOS Command Line Interface

```
Password:

S_CPT_Printers>en
Password:
S_CPT_Printers#show inter
S_CPT_Printers#show interfaces stt
S_CPT_Printers#show interfaces st
S_CPT_Printers#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    60         auto    auto  10/100BaseTX
Fa0/2                        connected    60         auto    auto  10/100BaseTX
Fa0/3                        notconnect   60         auto    auto  10/100BaseTX
Fa0/4                        notconnect   60         auto    auto  10/100BaseTX
Fa0/5                        notconnect   60         auto    auto  10/100BaseTX
Fa0/6                        notconnect   60         auto    auto  10/100BaseTX
Fa0/7                        notconnect   60         auto    auto  10/100BaseTX
Fa0/8                        notconnect   60         auto    auto  10/100BaseTX
Fa0/9                        notconnect   60         auto    auto  10/100BaseTX
Fa0/10                       notconnect   60         auto    auto  10/100BaseTX
Fa0/11                       notconnect   60         auto    auto  10/100BaseTX
Fa0/12                       notconnect   60         auto    auto  10/100BaseTX
Fa0/13                       notconnect   60         auto    auto  10/100BaseTX
Fa0/14                       notconnect   60         auto    auto  10/100BaseTX
Fa0/15                       notconnect   60         auto    auto  10/100BaseTX
Fa0/16                       notconnect   60         auto    auto  10/100BaseTX
Fa0/17                       notconnect   60         auto    auto  10/100BaseTX
Fa0/18                       notconnect   60         auto    auto  10/100BaseTX
Fa0/19                       notconnect   60         auto    auto  10/100BaseTX
Fa0/20                       notconnect   60         auto    auto  10/100BaseTX
Fa0/21                       notconnect   60         auto    auto  10/100BaseTX
Fa0/22                       notconnect   60         auto    auto  10/100BaseTX
Fa0/23                       notconnect   60         auto    auto  10/100BaseTX
Fa0/24                       connected    trunk      auto    auto  10/100BaseTX
Gig0/1                       connected    trunk      auto    auto  10/100BaseTX
Gig0/2                       connected    trunk      auto    auto  10/100BaseTX

S_CPT_Printers#
```

Copy Paste

Top

CPT_Management

Physical Config CLI Attributes

IOS Command Line Interface

```
Username: cisco
Password:

S_CPT_Management>en
Password:
S_CPT_Management#show interf
S_CPT_Management#show interfaces st
S_CPT_Management#show interfaces status
Port      Name                Status       Vlan       Duplex  Speed Type
Fa0/1                         connected    99         auto    auto  10/100BaseTX
Fa0/2                         notconnect   99         auto    auto  10/100BaseTX
Fa0/3                         notconnect   99         auto    auto  10/100BaseTX
Fa0/4                         notconnect   99         auto    auto  10/100BaseTX
Fa0/5                         notconnect   99         auto    auto  10/100BaseTX
Fa0/6                         connected    99         auto    auto  10/100BaseTX
Fa0/7                         notconnect   99         auto    auto  10/100BaseTX
Fa0/8                         notconnect   99         auto    auto  10/100BaseTX
Fa0/9                         notconnect   99         auto    auto  10/100BaseTX
Fa0/10                        notconnect   99         auto    auto  10/100BaseTX
Fa0/11                        notconnect   99         auto    auto  10/100BaseTX
Fa0/12                        notconnect   99         auto    auto  10/100BaseTX
Fa0/13                        notconnect   99         auto    auto  10/100BaseTX
Fa0/14                        notconnect   99         auto    auto  10/100BaseTX
Fa0/15                        notconnect   99         auto    auto  10/100BaseTX
Fa0/16                        notconnect   99         auto    auto  10/100BaseTX
Fa0/17                        notconnect   99         auto    auto  10/100BaseTX
Fa0/18                        notconnect   99         auto    auto  10/100BaseTX
Fa0/19                        notconnect   99         auto    auto  10/100BaseTX
Fa0/20                        notconnect   99         auto    auto  10/100BaseTX
Fa0/21                        notconnect   99         auto    auto  10/100BaseTX
Fa0/22                        notconnect   99         auto    auto  10/100BaseTX
Fa0/23                        notconnect   99         auto    auto  10/100BaseTX
Fa0/24                        connected    trunk      auto    auto  10/100BaseTX
Gig0/1                        connected    trunk      auto    auto  10/100BaseTX
Gig0/2                        connected    trunk      auto    auto  10/100BaseTX

S_CPT_Management#
```

Copy Paste

Top

For Potchefstroom network devices:

POT_Hosts

Physical Config CLI Attributes

IOS Command Line Interface

```
Username: cisco
Password:

S_POT_Hosts>en
Password:
S_POT_Hosts#show interface sta
S_POT_Hosts#show interface status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    40         auto    auto  10/100BaseTX
Fa0/2                        connected    40         auto    auto  10/100BaseTX
Fa0/3                        connected    40         auto    auto  10/100BaseTX
Fa0/4                        connected    40         auto    auto  10/100BaseTX
Fa0/5                        connected    40         auto    auto  10/100BaseTX
Fa0/6                        connected    40         auto    auto  10/100BaseTX
Fa0/7                        connected    40         auto    auto  10/100BaseTX
Fa0/8                        connected    40         auto    auto  10/100BaseTX
Fa0/9                        connected    40         auto    auto  10/100BaseTX
Fa0/10                       connected    40         auto    auto  10/100BaseTX
Fa0/11                       connected    40         auto    auto  10/100BaseTX
Fa0/12                       connected    40         auto    auto  10/100BaseTX
Fa0/13                       connected    40         auto    auto  10/100BaseTX
Fa0/14                       connected    40         auto    auto  10/100BaseTX
Fa0/15                       connected    40         auto    auto  10/100BaseTX
Fa0/16                       connected    40         auto    auto  10/100BaseTX
Fa0/17                       connected    40         auto    auto  10/100BaseTX
Fa0/18                       connected    40         auto    auto  10/100BaseTX
Fa0/19                       connected    40         auto    auto  10/100BaseTX
Fa0/20                       connected    40         auto    auto  10/100BaseTX
Fa0/21                       notconnect   40         auto    auto  10/100BaseTX
Fa0/22                       notconnect   40         auto    auto  10/100BaseTX
Fa0/23                       notconnect   40         auto    auto  10/100BaseTX
Fa0/24                       connected    trunk      auto    auto  10/100BaseTX
Gig0/1                       connected    trunk      auto    auto  10/100BaseTX
Gig0/2                       connected    trunk      auto    auto  10/100BaseTX

S_POT_Hosts#
```

Copy Paste

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pot_Management(1)

Physical Config CLI Attributes

IOS Command Line Interface

```
Username: cisco
Password:

S_POT_Management>en
Password:
S_POT_Management#show inter
S_POT_Management#show interfaces st
S_POT_Management#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    99         auto    auto  10/100BaseTX
Fa0/2                        notconnect   99         auto    auto  10/100BaseTX
Fa0/3                        notconnect   99         auto    auto  10/100BaseTX
Fa0/4                        notconnect   99         auto    auto  10/100BaseTX
Fa0/5                        notconnect   99         auto    auto  10/100BaseTX
Fa0/6                        connected    99         auto    auto  10/100BaseTX
Fa0/7                        notconnect   99         auto    auto  10/100BaseTX
Fa0/8                        notconnect   99         auto    auto  10/100BaseTX
Fa0/9                        notconnect   99         auto    auto  10/100BaseTX
Fa0/10                       notconnect   99         auto    auto  10/100BaseTX
Fa0/11                       notconnect   99         auto    auto  10/100BaseTX
Fa0/12                       notconnect   99         auto    auto  10/100BaseTX
Fa0/13                       notconnect   99         auto    auto  10/100BaseTX
Fa0/14                       notconnect   99         auto    auto  10/100BaseTX
Fa0/15                       notconnect   99         auto    auto  10/100BaseTX
Fa0/16                       notconnect   99         auto    auto  10/100BaseTX
Fa0/17                       notconnect   99         auto    auto  10/100BaseTX
Fa0/18                       notconnect   99         auto    auto  10/100BaseTX
Fa0/19                       notconnect   99         auto    auto  10/100BaseTX
Fa0/20                       notconnect   99         auto    auto  10/100BaseTX
Fa0/21                       notconnect   99         auto    auto  10/100BaseTX
Fa0/22                       notconnect   99         auto    auto  10/100BaseTX
Fa0/23                       notconnect   99         auto    auto  10/100BaseTX
Fa0/24                       connected    trunk      auto    auto  10/100BaseTX
Gig0/1                       connected    trunk      auto    auto  10/100BaseTX
Gig0/2                       connected    trunk      auto    auto  10/100BaseTX

S_POT_Management#
```

Copy Paste

Top

POT_Printers(1)

Physical Config CLI Attributes

IOS Command Line Interface

```
Username: cisco
Password:

S_POT_Printers>en
Password:
S_POT_Printers#show inter
S_POT_Printers#show interfaces st
S_POT_Printers#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    60         auto    auto  10/100BaseTX
Fa0/2                        connected    60         auto    auto  10/100BaseTX
Fa0/3                        notconnect   60         auto    auto  10/100BaseTX
Fa0/4                        notconnect   60         auto    auto  10/100BaseTX
Fa0/5                        notconnect   60         auto    auto  10/100BaseTX
Fa0/6                        notconnect   60         auto    auto  10/100BaseTX
Fa0/7                        notconnect   60         auto    auto  10/100BaseTX
Fa0/8                        notconnect   60         auto    auto  10/100BaseTX
Fa0/9                        notconnect   60         auto    auto  10/100BaseTX
Fa0/10                       notconnect   60         auto    auto  10/100BaseTX
Fa0/11                       notconnect   60         auto    auto  10/100BaseTX
Fa0/12                       notconnect   60         auto    auto  10/100BaseTX
Fa0/13                       notconnect   60         auto    auto  10/100BaseTX
Fa0/14                       notconnect   60         auto    auto  10/100BaseTX
Fa0/15                       notconnect   60         auto    auto  10/100BaseTX
Fa0/16                       notconnect   60         auto    auto  10/100BaseTX
Fa0/17                       notconnect   60         auto    auto  10/100BaseTX
Fa0/18                       notconnect   60         auto    auto  10/100BaseTX
Fa0/19                       notconnect   60         auto    auto  10/100BaseTX
Fa0/20                       notconnect   60         auto    auto  10/100BaseTX
Fa0/21                       notconnect   60         auto    auto  10/100BaseTX
Fa0/22                       notconnect   60         auto    auto  10/100BaseTX
Fa0/23                       notconnect   60         auto    auto  10/100BaseTX
Fa0/24                       connected    trunk      auto    auto  10/100BaseTX
Gig0/1                       connected    trunk      auto    auto  10/100BaseTX
Gig0/2                       connected    trunk      auto    auto  10/100BaseTX

S_POT_Printers#
```

Copy Paste

Top

Switch2

```
S_POT#show int
S_POT#show interfaces st
Port      Name               Status       Vlan       Duplex  Speed Type
Gig5/1                       connected    trunk      auto    auto  10/100BaseTX
Gig6/1                       notconnect   1          auto    auto  10/100BaseTX
Gig7/1                       connected    trunk      auto    auto  10/100BaseTX
Gig8/1                       connected    trunk      auto    auto  10/100BaseTX
Gig9/1                       connected    trunk      auto    auto  10/100BaseTX
```

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OSPF(Open Shortest Path First) is a link-state routing protocol used to find the best path between routers within an Autonomous Systems as defined by (GeekforGeeks, 2025).Below you'll find the configuration of OSPF for GlobeTech

R1_JHB

Physical Config CLI Attributes

IOS Command Line Interface

```
R1_JHB(config-if)#exit
R1_JHB(config)#exit
R1_JHB#
%SYS-5-CONFIG_I: Configured from console by console

R1_JHB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1_JHB(config)#interface lo
R1_JHB(config)#interface loopback0

R1_JHB(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R1_JHB(config-if)#ip address 10.255.0.1 255.255.255.255
R1_JHB(config-if)#no shutdown
R1_JHB(config-if)#exit
R1_JHB(config)#router ospf 1
R1_JHB(config-router)#router-id 10.255.9.1
R1_JHB(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1_JHB(config-router)#network 10.0.0.8 0.0.0.3 area 0
R1_JHB(config-router)#network 10.0.0.12 0.0.0.3 area 0
R1_JHB(config-router)#network 10.0.0.16 0.0.0.3 area 0
R1_JHB(config-router)#network 10.0.0.24 0.0.0.3 area 0
R1_JHB(config-router)#network 10.0.0.28 0.0.0.3 area 0
R1_JHB(config-router)#network 10.255.0.1 0.0.0.0 area 0
R1_JHB(config-router)#exit
R1_JHB(config)#exit
R1_JHB#
%SYS-5-CONFIG_I: Configured from console by console

R1_JHB#w m
Building configuration...
[OK]
R1_JHB#
```

Copy Paste

Top

```
R1_JHB#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.0.12       1   FULL/BDR        00:00:30    10.0.0.26       GigabitEthernet0/0/1
10.255.0.5        0   FULL/  -        00:00:37    10.0.0.2        Serial0/1/1
10.255.0.3        0   FULL/  -        00:00:30    10.0.0.14       Serial0/1/0
R1_JHB#
```

Copy Paste

Top

R2_CPT

Physical Config CLI Attributes

IOS Command Line Interface

```
R2_CPT(config)#interface lo
R2_CPT(config)#interface loopback0

R2_CPT(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R2_CPT(config-if)#ip address 10.255.0.5 255.255.255.255
R2_CPT(config-if)#no shutdown
R2_CPT(config-if)#exiit
                     ^
% Invalid input detected at '^' marker.

R2_CPT(config-if)#exit
R2_CPT(config)#router ospf 1
R2_CPT(config-router)#router-id 10.255.0.5
R2_CPT(config-router)#network 10.0.0.0 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.4 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.8 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.12 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.16 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.10 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.24 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.28 0.0.0.3 area 0
R2_CPT(config-router)#no network 10.0.0.10 0.0.0.3 area 0
R2_CPT(config-router)#network 10.0.0.20 0.0.0.3 area 0
R2_CPT(config-router)#network 10.255.0.5 0.0.0.0 area 0
R2_CPT(config-router)#exit
R2_CPT(config)#exit
R2_CPT#
%SYS-5-CONFIG_I: Configured from console by console

R2_CPT#w m
Building configuration...
[OK]
R2_CPT#
```

Copy Paste

Top

```
R2_CPT#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.0.11       1   FULL/BDR        00:00:33    10.0.0.22       GigabitEthernet0/0/0
10.255.9.1        0   FULL/  -        00:00:34    10.0.0.1        Serial0/1/0
R2_CPT#
```

R3_POT

Physical Config CLI Attributes

IOS Command Line Interface

```
R3_POT(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R3_POT(config-if)#ip address 10.255.0.3 255.255.255.255
R3_POT(config-if)#no shutdown
R3_POT(config-if)#exit
R3_POT(config)#router ospf 2
R3_POT(config-router)#exit
R3_POT(config)#router ospf 1
R3_POT(config-router)#router-id 10.255.0.3
OSPF: router-id 10.255.0.3 in use by ospf process 2
R3_POT(config-router)#exit
R3_POT(config)#no router ospf 2
R3_POT(config)#router ospf 1
R3_POT(config-router)#router-id 10.255.0.3
R3_POT(config-router)#network 10.0.0.0R3_POT(config-router)#
R3_POT(config-router)#end
R3_POT#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3_POT(config)#interface Serial0/1/0
R3_POT(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

R3_POT(config-if)#exit
R3_POT(config)#router ospf 1
R3_POT(config-router)#network 10.0.0.12 0.0.0.3 area 0
R3_POT(config-router)#
06:15:12: %OSPF-5-ADJCHG: Process 1, Nbr 10.255.9.1 on Serial0/1/0 from LOADING to FULL,
Loading Done

R3_POT(config-router)#network 10.2255.0.3 0.0.0.0 area 0
                                ^
% Invalid input detected at '^' marker.

R3_POT(config-router)#network 10.255.0.3 0.0.0.0 area 0
```

Copy Paste

Top

```
R3_POT#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.9.1        0   FULL/  -        00:00:30    10.0.0.13       Serial0/1/0
R3_POT#
```

Multilayer Switch_JHB

Physical Config CLI Attributes

IOS Command Line Interface

```
MS1_JHB>en
Password:
MS1_JHB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MS1_JHB(config)#interface loo
MS1_JHB(config)#interface loopback0

MS1_JHB(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

MS1_JHB(config-if)#ip address 10.255.0.11 255.255.255.255
MS1_JHB(config-if)#no shurtdown
                   ^
% Invalid input detected at '^' marker.

MS1_JHB(config-if)#no shutdown
MS1_JHB(config-if)#exit
MS1_JHB(config)#router ospf 1
MS1_JHB(config-router)#router-id 10.255.0.11
MS1_JHB(config-router)#network 10.0.0.20 0.0.0.3 area 0
MS1_JHB(config-router)#network
06:27:50: %OSPF-5-ADJCHG: Process 1, Nbr 10.255.0.5 on GigabitEthernet1/0/2 from
LOADING to FULL, Loading Done

% Incomplete command.
MS1_JHB(config-router)#netowork 10.255.0.11 0.0.0.0 area 0
                      ^
% Invalid input detected at '^' marker.

MS1_JHB(config-router)#network 10.255.0.11 0.0.0.0 area 0
MS1_JHB(config-router)#exit
MS1_JHB(config)#exit
MS1_JHB#
%SYS-5-CONFIG_I: Configured from console by console
w m
```

Copy Paste

Top

```
MS1_JHB(config-router)#network 172.16.3.128 0.0.0.15 area 0
MS1_JHB(config-router)#network 172.16.2.0 0.0.0.127 area 0
MS1_JHB(config-router)#network 172.16.3.96 0.0.0.31  area 0
MS1_JHB(config-router)#network 172.16.2.128 0.0.0.127 area 0
MS1_JHB(config-router)#network 172.16.0.0 0.0.1.255 area 0
MS1_JHB(config-router)#exit
MS1_JHB(config)#exit
MS1_JHB#
%SYS-5-CONFIG_I: Configured from console by console
```

```
MS1_JHB#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.0.5        1   FULL/DR         00:00:30    10.0.0.21       GigabitEthernet1/0/2
MS1_JHB#
```

Multilayer Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
MS2_CPT#w m
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
MS2_CPT#show ip ospf neighbor

MS2_CPT#show ip ospf neighbor

MS2_CPT#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
MS2_CPT(config)#router ospf 1
MS2_CPT(config-router)#no network 10.0.0.24 0.0.0.0 area 0
MS2_CPT(config-router)#network 10.0.0.24 0.0.0.3 area 0
MS2_CPT(config-router)#
06:35:33: %OSPF-5-ADJCHG: Process 1, Nbr 10.255.9.1 on GigabitEthernet1/0/2 from
LOADING to FULL, Loading Done

MS2_CPT(config-router)#
MS2_CPT(config-router)#exit
MS2_CPT(config)#exit
MS2_CPT#
%SYS-5-CONFIG_I: Configured from console by console
wm
Translating "wm"
% Unknown command or computer name, or unable to find computer address

MS2_CPT#w m
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
MS2_CPT#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.9.1        1   FULL/DR         00:00:30    10.0.0.25       GigabitEthernet1/0/2
MS2_CPT#
```

Copy Paste

Top

```
MS2_CPT(config-router)#network 172.16.5.128 0.0.0.15 area 0
MS2_CPT(config-router)#network 172.16.3.0 0.0.0.63 area 0
MS2_CPT(config-router)#network 172.16.5.96 0.0.0.31 area 0
MS2_CPT(config-router)#network 172.16.7.128 0.0.0.127 area 0
MS2_CPT(config-router)#network 172.16.4.0 0.0.0.127 area 0
MS2_CPT(config-router)#exit
MS2_CPT(config)#exit
MS2_CPT#
%SYS-5-CONFIG_I: Configured from console by console

MS2_CPT#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.9.1        1   FULL/DR         00:00:30    10.0.0.25       GigabitEthernet1/0/2
MS2_CPT#
```

ACLs (Access Lists) :are a list of rules that determine which user/system as access to which resources. Below you'll find configured ACLs to limit Guest-Wi-Fi and only allow IT/management access to network devices:

Guest-WI-FI limiting in:

Johannesburg:

```
MS1_JHB>en
Password:
MS1_JHB#config int
MS1_JHB#config t
Enter configuration commands, one per line.  End with CNTL/Z.
MS1_JHB(config)#ip access-list extended GUEST-INTERNET-ONLY
MS1_JHB(config-ext-nacl)#permit udp 172.16.2.128 0.0.0.127 any eq 67
MS1_JHB(config-ext-nacl)#permit udp any eq 67 172.16.2.128 0.0.0.127 any eq 68
                                                                      ^
% Invalid input detected at '^' marker.

MS1_JHB(config-ext-nacl)#permit udp any eq 67 172.16.2.128 0.0.0.127 eq 68
MS1_JHB(config-ext-nacl)#permit ip 172.16.2.128 0.0.0.127 any
MS1_JHB(config-ext-nacl)#exit
MS1_JHB(config)#interface vlan 100
MS1_JHB(config-if)#ip access-group GUEST-INTERNET-ONLY in
MS1_JHB(config-if)#
```

ACLs edited to allow WLC discovery for AP

```
MS1_JHB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MS1_JHB(config)#ip access-list extended GUEST-INTERNET-ONLY
MS1_JHB(config-ext-nacl)#permit udp 172.16.2.128 0.0.0.127 172.16.3.128 0.0.0.15
range 5246 5247
MS1_JHB(config-ext-nacl)#exit
MS1_JHB(config)#interface vlan 100
MS1_JHB(config-if)#ip ac
MS1_JHB(config-if)#ip access-group GUEST-INTERNET-ONLY
% Incomplete command.
MS1_JHB(config-if)#ip access-group GUEST-INTERNET-ONLY IN
MS1_JHB(config-if)#EXIT
MS1_JHB(config)#
```

Cape Town:

```
MS2_CPT>en
Password:
MS2_CPT#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MS2_CPT(config)#ip access-list extended GUEST-INTERNET-ONLY
MS2_CPT(config-ext-nacl)#permit udp any 172.16.7.128 0.0.0.127 any eq 67
                                                                ^
% Invalid input detected at '^' marker.

MS2_CPT(config-ext-nacl)#permit udp 172.16.7.128 0.0.0.127 any eq 67
MS2_CPT(config-ext-nacl)#permit udp any eq 67 172.16.7.128 0.0.0.127 eq 68
MS2_CPT(config-ext-nacl)#permit ip 172.16.7.128 0.0.0.127 any
MS2_CPT(config-ext-nacl)#exit
MS2_CPT(config)#interf
MS2_CPT(config)#interface vlan 100
MS2_CPT(config-if)#ip access-group GUEST-INTERNET-ONLY in
MS2_CPT(config-if)#exit
MS2_CPT(config)#exit
MS2_CPT#
%SYS-5-CONFIG_I: Configured from console by console

MS2_CPT#w m
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
MS2_CPT#
```

ACLs edited to allow WLC discovery for AP

NAT (Network Address Translation) allows multiple devices using varying private IP addresses to use the same public address. Below you find the results of nat configuration on edge routers:

R1_JHB:

```
R1_JHB#show run | section interface
interface Loopback0
 ip address 10.255.0.1 255.255.255.255
interface GigabitEthernet0/0/0
 ip address 10.0.0.17 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
interface GigabitEthernet0/0/1
 ip address 10.0.0.25 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
interface Serial0/1/0
 ip address 10.0.0.13 255.255.255.252
 ip nat inside
interface Serial0/1/1
 ip address 10.0.0.1 255.255.255.252
 ip nat inside
interface Serial0/2/0
 ip address 10.0.0.5 255.255.255.252
 ip nat outside
 clock rate 2000000
```

```
R1_JHB(config)#access-list 1 permit 172.16.0.0 0.0.255.255
R1_JHB(config)#ip nat inside source list 1 interface serial0/2/0 overload
R1_JHB(config)#
```

R2_CPT

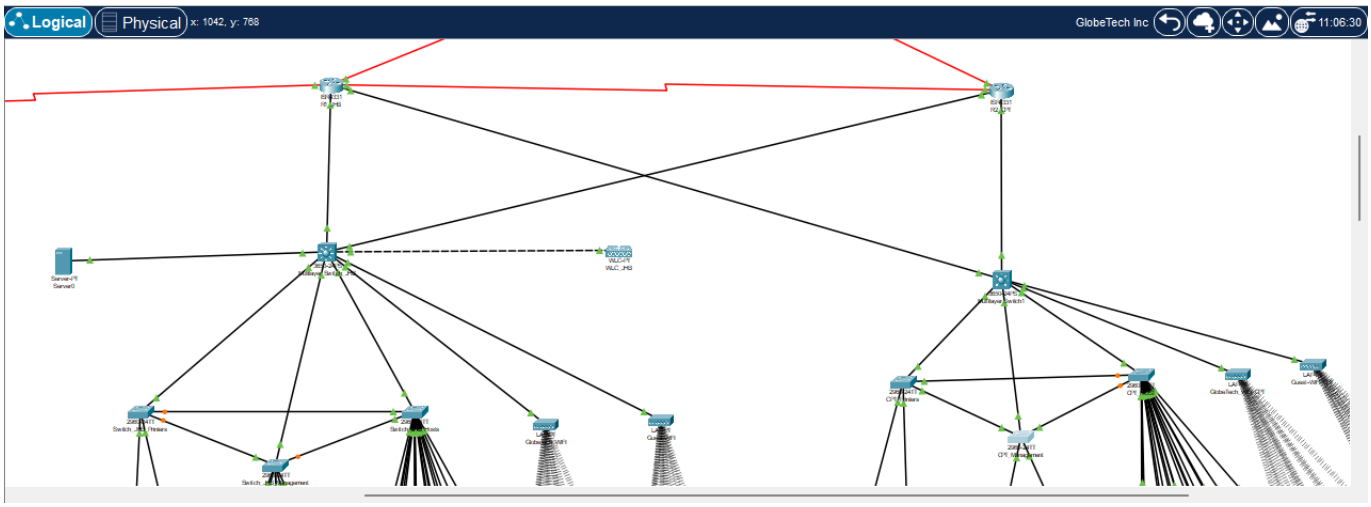
```
R2_CPT#show run | section interface
interface Loopback0
 ip address 10.255.0.5 255.255.255.255
interface GigabitEthernet0/0/0
 ip address 10.0.0.21 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
interface GigabitEthernet0/0/1
 ip address 10.0.0.29 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
interface Serial0/1/0
 ip address 10.0.0.2 255.255.255.252
 ip nat inside
 clock rate 2000000
interface Serial0/1/1
 ip address 10.0.0.9 255.255.255.252
 ip nat outside
 clock rate 2000000
interface Vlan1
 no ip address
 shutdown
ip nat inside source list 1 interface Serial0/1/1 overload
R2_CPT(config)#access-list 1 permit 172.16.0.0 0.0.255.255
R2_CPT(config)#ip nat inside source list 1 interface serial0/1/1
```

HSRP (Hot Standby Router Protocol) is a redundancy protocol for setting up a fault-tolerant default gateway in a LAN environment, as defined by (Packet Tracer Network, 2024).Below you'll find the HSRP configuration for GlobeTech: Unfortunately my topology does not support HSPR since my routers are not the same subnet due using /30 say my subnet mask

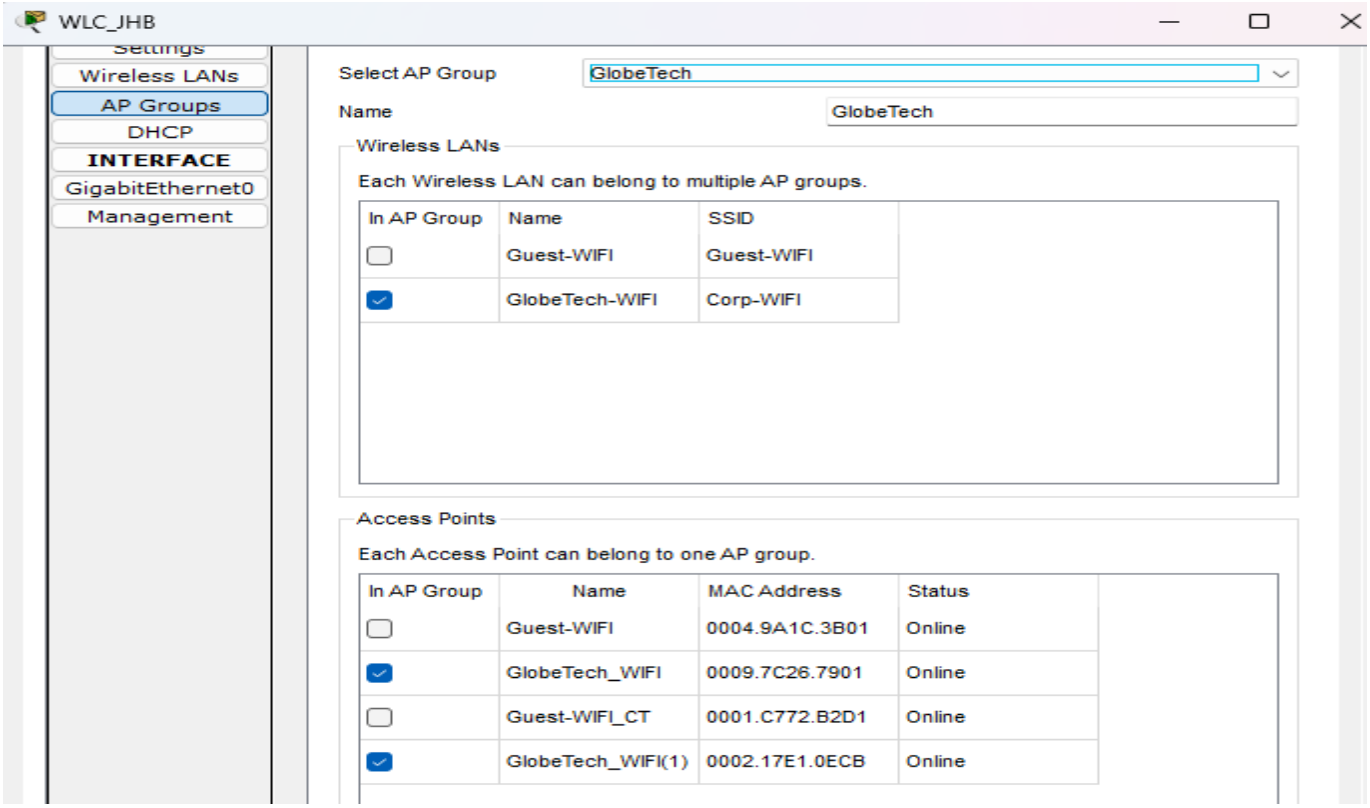
Wireless configuration

Wireless LAN Controller

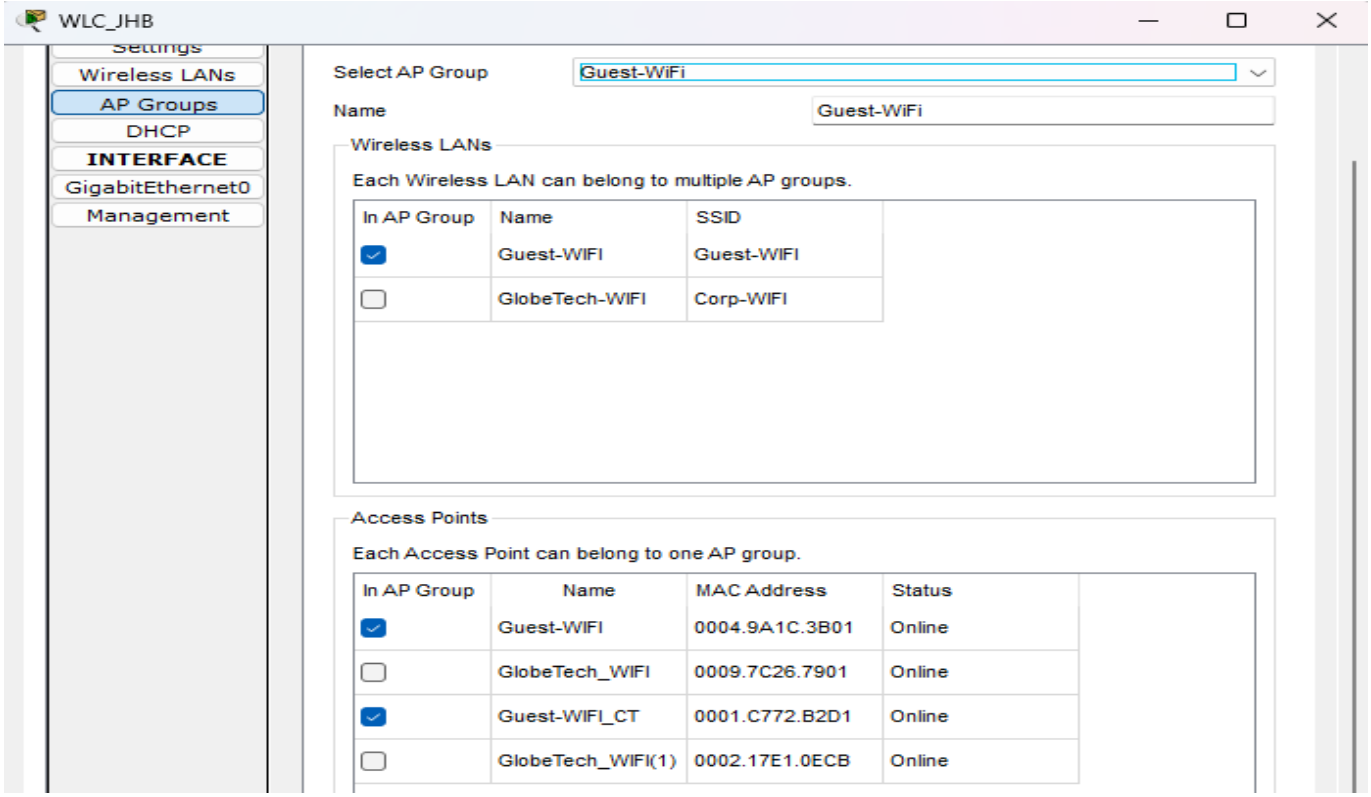
- Deployed WLC in Johannesburg and Cape Town
  - Central WLC for both sites:



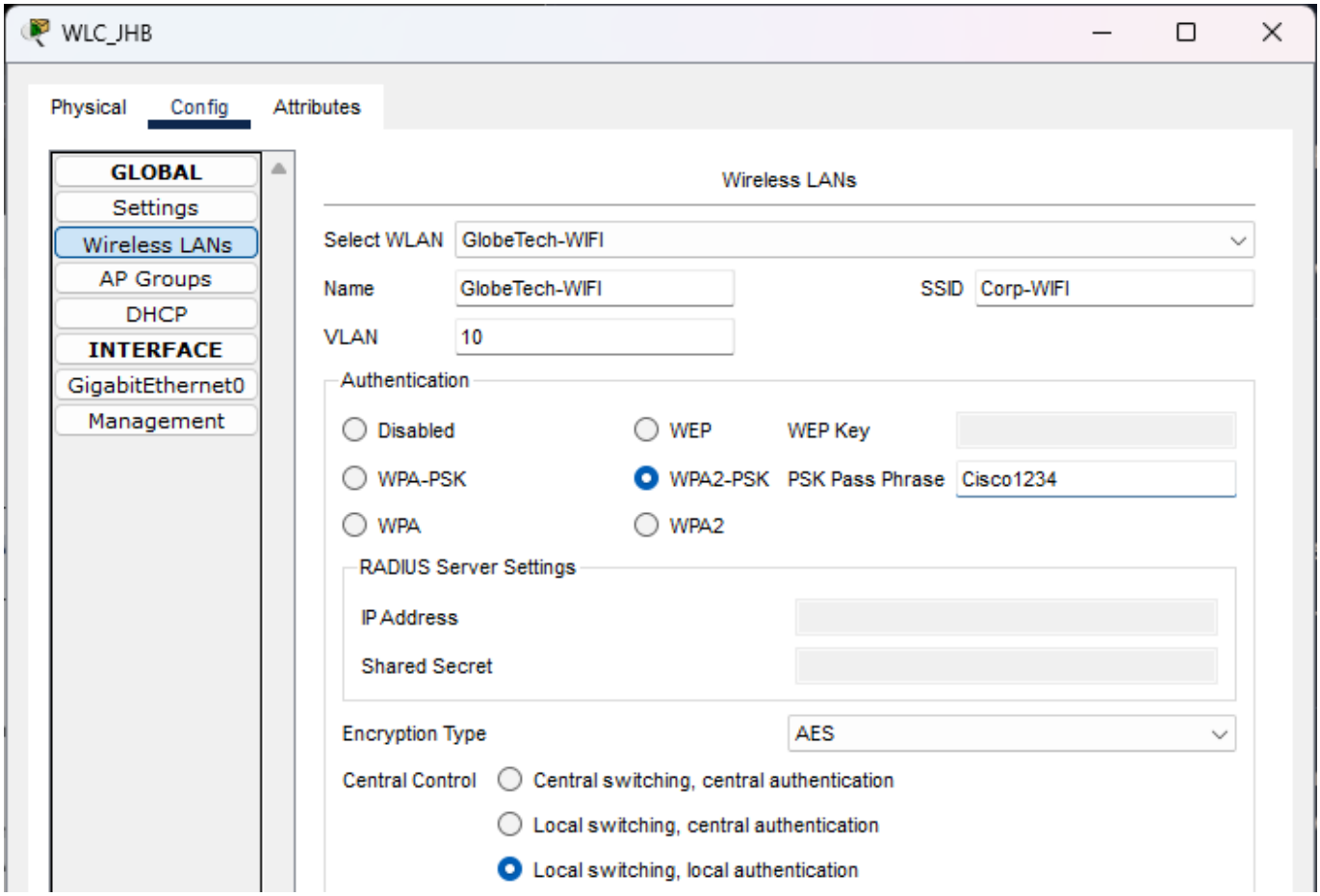
- Configured and registered 2 APs per Site
  - GlobeTech-WIFI APs:



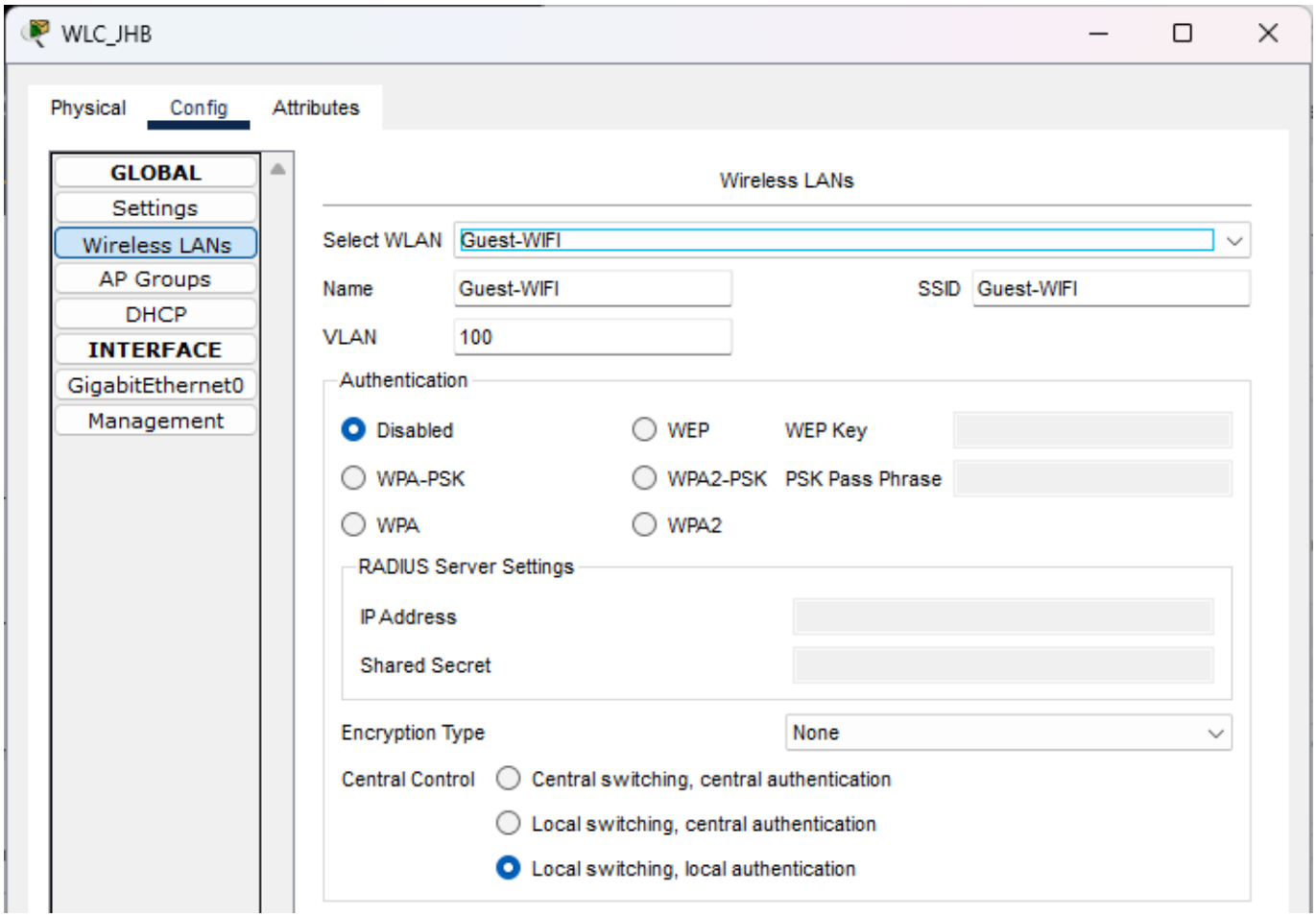
  - Guest-WIFI APs:



- Created 2 SSIDs:
  - Corp-WIFI (WPA2 secured (Password: Cisco12334), VLAN segmented (VLAN 10))



- Guest-WIFI (internet only, restricted by ACL)



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